# Teletext Navigator

User’s Guide

Version 1.6

For Microsoft Windows 10 and Windows 11
64-bit (x64)

## The Teletext

**Teletext Navigator User's Guide, version 1.6**
November 2026

Copyright © 2026 Dimitri Papadopoulos and The Teletext.

This manual is free documentation: you can redistribute it and modify it under the terms of the GNU Affero General Public License as published by the Free Software Foundation, version 3 of the License or (at your option) any later version, the same licence as the program it describes.

Teletext Navigator is free software: you can redistribute it and modify it under the terms of the GNU Affero General Public License as published by the Free Software Foundation, version 3 of the License or (at your option) any later version. The Teletext WinHelp (TTWINHLP.EXE), the Help viewer installed with it, is free software under the GNU Lesser General Public License, version 2.1 or later; its source code is at https://github.com/TheTeletext/WinHelp. The licences of the other software Teletext Navigator contains are listed in its About box and in the LICENSES folder of the program.

The Teletext, http://www.teletext.ca

Microsoft, Windows and GW-BASIC are trademarks of Microsoft Corporation. RIPscrip is a trademark of TeleGrafix Communications, Inc. Flash is a trademark of Adobe; Ruffle is independent of Adobe. RealAudio and RealPlayer are trademarks of RealNetworks, Inc. Other names are the trademarks of their owners.

This manual is set in Liberation Serif, Liberation Sans and Liberation Mono (SIL Open Font License 1.1) with Typst.

# Foreword

The Internet I grew up with did not begin with the World Wide Web. Before it, and for years beside it, there were bulletin boards drawn in RIPscrip, videotex pages on the Minitel and on Bell's ALEX, Gopher menus, WAIS searches, Archie, anonymous FTP, Netnews, Internet Relay Chat, and live television over CU-SeeMe. Each of these protocols carried news and ideas in its day, and most of them have been all but forgotten.

The purpose of this software is to preserve the legacy of these historically significant protocols, and to make them accessible on a modern computer. Every service in Teletext Navigator is the real thing: Navigator speaks each protocol the way its own clients did, to live servers that The Teletext keeps running today.

You will notice the look at once. Navigator wears the skin of Windows 3.11: the navy title bars, the grey buttons, the Help files. That skin is deliberate. It was chosen for immersion in the Old Net, just as it was back in the day. But don't be fooled by the looks: under the skin, Teletext Navigator is a native x64 application, running natively on Windows 11.

I hope you enjoy using it as much as I enjoyed architecting it.

**Dimitri Papadopoulos**

**The Teletext**

# Contents

**Foreword** ..... **i**

**1 Introduction** ..... **1**
- The Windows 3.1 Look and Feel ..... 2
- What You Need ..... 3
- Conventions Used in This Manual ..... 3

**2 Installing Teletext Navigator** ..... **4**
- What Is Installed ..... 4
- Starting and Updating ..... 4
- Removing Teletext Navigator ..... 4

**3 Getting Started** ..... **6**
- The Navigator Window ..... 6
- Choosing a Service: the Ribbon ..... 7
- Menu Commands ..... 7
- The Status Bar ..... 9
- Checking for Updates ..... 9
- Getting Help ..... 10

**4 Home** ..... **11**

**5 RIPscrip** ..... **12**
- Connecting to a RIPscrip Board ..... 12
- RIPscrip Settings ..... 13
- Using a RIPscrip Board ..... 13
- RIPscrip Troubleshooting ..... 14

**6 NABU** ..... **16**
- Starting the NABU ..... 16
- NABU Settings ..... 17
- Using the NABU ..... 17
- NABU Troubleshooting ..... 18

**7 Videotex** ..... **19**
- Connecting to a Videotex Service ..... 19
- Videotex Settings ..... 20
- Using Minitel and NAPLPS ..... 20
- Videotex Troubleshooting ..... 21

**8 Web** ..... **22**

Opening a Web Page . . . . . 22
Web Settings . . . . . 23
Browsing the Web . . . . . 23
Web Troubleshooting . . . . . 26

**9 Gopher . . . . . 28**
Opening a Gopher Menu . . . . . 28
Gopher Settings . . . . . 29
Using Gopher Menus . . . . . 29
Gopher Troubleshooting . . . . . 30

**10 ARPANET . . . . . 31**
Connecting to an FTP-Mail Server . . . . . 31
ARPANET Settings . . . . . 31
Writing and Sending a Message . . . . . 32
ARPANET Troubleshooting . . . . . 33

**11 WAIS . . . . . 34**
Connecting to a WAIS Server . . . . . 34
WAIS Settings . . . . . 35
Searching and Reading Documents . . . . . 35
WAIS Troubleshooting . . . . . 37

**12 Archie . . . . . 39**
Searching an Archie Server . . . . . 39
Archie Settings . . . . . 40
Choosing and Downloading a File . . . . . 40
Archie Troubleshooting . . . . . 41

**13 FTP . . . . . 42**
Connecting to an FTP Server . . . . . 42
FTP Settings . . . . . 43
Browsing and Downloading Files . . . . . 43
FTP Troubleshooting . . . . . 44

**14 Newsgroups . . . . . 46**
Connecting to a News Server . . . . . 46
Newsgroups Settings . . . . . 47
Reading Newsgroups . . . . . 47
Newsgroups Troubleshooting . . . . . 48

**15 IRC . . . . . 49**
Connecting to an IRC Channel . . . . . 49
IRC Settings . . . . . 49
Reading the Channel . . . . . 50
IRC Troubleshooting . . . . . 51

**16 CU-SeeMe . . . . . 52**
Connecting to a Reflector . . . . . 52
CU-SeeMe Settings . . . . . 53
Watching CU-SeeMe . . . . . 53
CU-SeeMe Troubleshooting . . . . . 54

**17 Terminal ..... 55**
Connecting from the Terminal ..... 55
Terminal Commands ..... 56
Terminal Settings ..... 57
Using the Terminal ..... 57
Terminal Troubleshooting ..... 58

**18 GW-BASIC ..... 60**
Starting GW-BASIC ..... 60
GW-BASIC Files and Settings ..... 60
Using GW-BASIC ..... 61
GW-BASIC Troubleshooting ..... 62

**19 Using Help ..... 63**
How to Use Help ..... 64
Jumps and Definitions ..... 64
The Help Buttons ..... 65
Searching for a Topic ..... 65
Going Back to Earlier Topics ..... 66
The Help Window ..... 66
Copying Help Text ..... 66
Keeping Help on Top ..... 66
Opening Another Help File ..... 66
Help Menu Commands ..... 66
About The Teletext WinHelp ..... 67

**Glossary ..... 68**

**Index ..... 71**

CHAPTER 1

# Introduction

Teletext Navigator connects your computer to the information services of The Teletext the way the networks of the 1980s and 1990s reached them. Each service has its own button on the ribbon:

- **RIPscrip**, a graphical bulletin board drawn in RIPscrip 1.54.
- **NABU**, the Canadian cable computer network of 1983, on an emulated NABU Personal Computer or on a real one.
- **Videotex**, the Minitel and NAPLPS videotex services.
- **Web**, a World Wide Web browser of the 1990s, with Flash.
- **Gopher**, the menus of Gopherspace.
- **ARPANET**, mail sent the FTP-Mail way.
- **WAIS**, full-text search in Z39.50-1988 (Version 1).
- **Archie**, the first Internet search engine, created in Montreal.
- **FTP**, anonymous file downloads.
- **Newsgroups**, a Netnews reader.
- **IRC**, the #retro channel as it happens.
- **CU-SeeMe**, live television over the Internet.
- **Terminal**, Telnet, Finger and QOTD.
- **GW-BASIC**, Microsoft's BASIC on a built-in PC.

Each service is set up for The Teletext's own server, so you can connect at once; most of them also let you name another host.

## The Windows 3.1 Look and Feel

Teletext Navigator wears the look of Windows 3.11 on purpose. Its navy title bars, its grey push buttons, its menus with their black borders and its dialog boxes were drawn pixel by pixel against the real Windows 3.11, running side by side, so that the services of the Old Net can be used in the very setting they were made for. The look is only skin deep. Underneath it, Teletext Navigator is a **native 64-bit (x64) application** for Windows 11 and Windows 10. It is not an old program running in an emulator, and it needs no emulator to run: every protocol is spoken by Navigator's own code, directly over your computer's Internet connection.

Two services do contain emulated machines, because their programs were written for a computer, not for a screen: the **NABU** service runs a NABU Personal Computer, and the **GW-BASIC** service runs an IBM PC compatible, so that the programs written for them run as they did. Navigator itself remains a native program around them.

**WinHelp, brought back.** Microsoft no longer includes WinHelp, the Help system of Windows 3.1, in Windows 10 and Windows 11. The Teletext went out of its way to bring it back. Navigator's Help is a real WinHelp system: its Help files are authentic .HLP files, compiled with the original Microsoft Help Compiler of 1993, and they are shown by **The Teletext WinHelp**, The Teletext's open-source revival of the WinHelp viewer. Based on the WinHelp viewer of the Wine project, it is faithful to WinHelp 3.1 down to the Search dialog box and the shadows under its pop-up definitions, and it opens the Help files of other Windows 3.0 and 3.1 programs as well. Its source code is at https://github.com/TheTeletext/WinHelp. See Using Help (page 63).

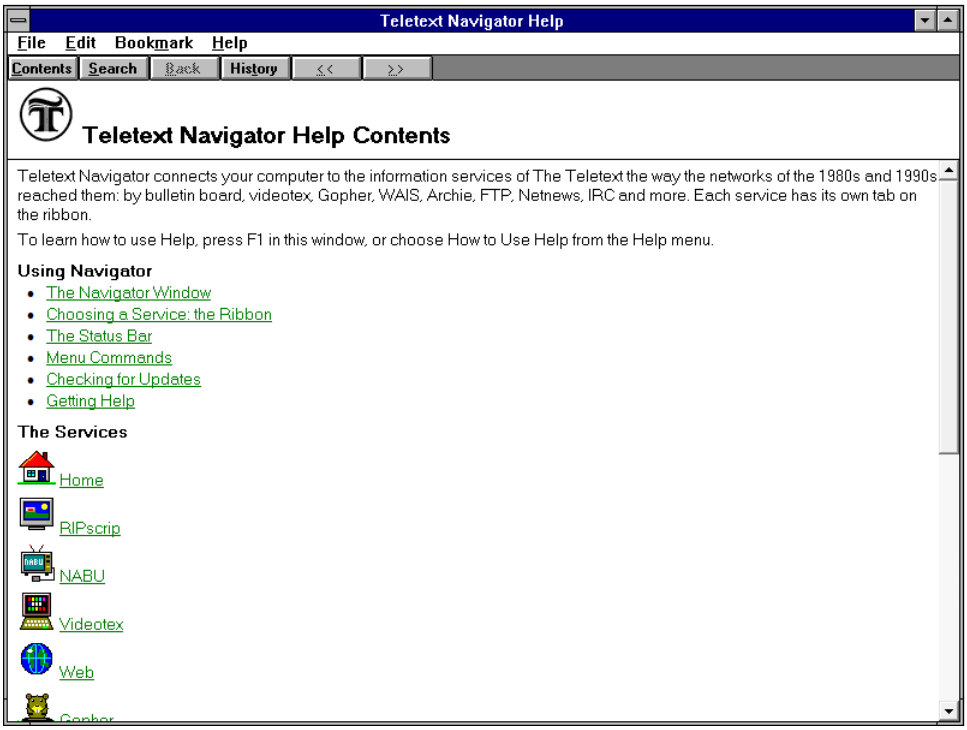



**Figure 1** Teletext Navigator Help at its Contents: a real WinHelp file in The Teletext WinHelp.

## What You Need

- A computer running a 64-bit edition of Windows. Teletext Navigator is tested on 64-bit Windows 10 22H2 and later, including Windows 11. Earlier versions are untested.
- A connection to the Internet. Every service except GW-BASIC reaches a server on the Internet.
- A sound device, for the sound of CU-SeeMe, NABU, GW-BASIC and the sounds in Gopher and the Web.
- A screen of at least 1024 by 768 pixels. The Navigator window is never smaller than that.
- For RealAudio links in the Web service, RealPlayer or another program that Windows uses for .ram files.

## Conventions Used in This Manual

- **Bold** text names something on the screen that you choose: a button, a box, a menu command or a list.
- Names of keys are in capital letters, for example ENTER, ESC and F1.
- A plus sign between key names, as in CTRL+C, means: hold down the first key while you press the second.
- "Choose" means to click an item with the mouse, or to select it with the keyboard.
- Addresses of services are given by their names, such as wais.teletext.ca, never by number.
- The pictures show Teletext Navigator connected to The Teletext's live services; what you see when you connect will be that day's news.

CHAPTER 2

# Installing Teletext Navigator

Teletext Navigator is installed with a Windows Installer package, a file whose name ends in .msi.

1. Open the installer file.
2. Read the licence agreement. Teletext Navigator is free software under the GNU Affero General Public License, version 3 or later. To continue, accept the agreement and choose **Next**.
3. In **Custom Setup**, choose what to install: **Teletext Navigator**: the program, its Help, this manual and the Start Menu shortcuts. It is always installed. **Desktop shortcut**: a shortcut to Teletext Navigator on the desktop. To install somewhere other than the Program Files\Teletext Navigator folder, choose **Browse**.
4. Choose **Install**. Windows asks your permission to install a program for all users of the computer.
5. Choose **Finish**.

If Teletext Navigator or its Help is open, the installer asks you to close it first.

## What Is Installed

In the program's folder:

- **TeletextNavigator.exe**, the program.
- **TTWINHLP.EXE**, The Teletext WinHelp, the Help viewer, with its licence (COPYING.LIB) and its own Help file, TTWINHLP.HLP.
- **TELETEXT.HLP**, Teletext Navigator's Help.
- **Teletext Navigator Manual.pdf**, this manual.
- README.TXT, LICENSE.TXT, NOTICES.TXT and CREDITS.TXT, and the LICENSES folder with the licences of the software Navigator contains.
- The NABU folder, with the start-up firmware of the NABU service.

On the Start Menu: **Teletext Navigator**, and **Teletext Navigator Manual**, which opens this manual.

## Starting and Updating

To start Teletext Navigator, choose **Teletext Navigator** on the Start Menu, or its desktop shortcut.

Navigator checks for a new version by itself, at most once a day, and you can check at any time with **Check for Updates** on the Help menu. An update downloads the new installer, checks it, and installs it over the old version: your settings are kept. See Checking for Updates (page 9).

## Removing Teletext Navigator

Open **Settings** in Windows and choose **Apps**, then **Installed apps** (on Windows 10, **Apps and features**). Choose Teletext Navigator, then **Uninstall**.

Files that you made with Navigator are yours and are not removed: the files you downloaded (in your Downloads folder or wherever you saved them) and your GW-BASIC programs (in the Teletext BASIC folder of your Documents).

CHAPTER 3

# Getting Started

## The Navigator Window

The Navigator window has five parts, from top to bottom:

- The **title bar**, with the Control-menu box at the left and the Minimize and Maximize buttons at the right.
- The **menu bar**, with the File and Help menus.
- The **ribbon**, a row of buttons, one for each service. See Choosing a Service: the Ribbon (page 7).
- The **work area**, where the chosen service is shown.
- The **status bar**, which reports what the service is doing. See The Status Bar (page 9).

The window can be sized by dragging its border. It cannot be made smaller than 1024 by 768 pixels, or smaller than your screen if your screen is smaller than that. Navigator opens at that size each time it starts; it does not remember a size or position.

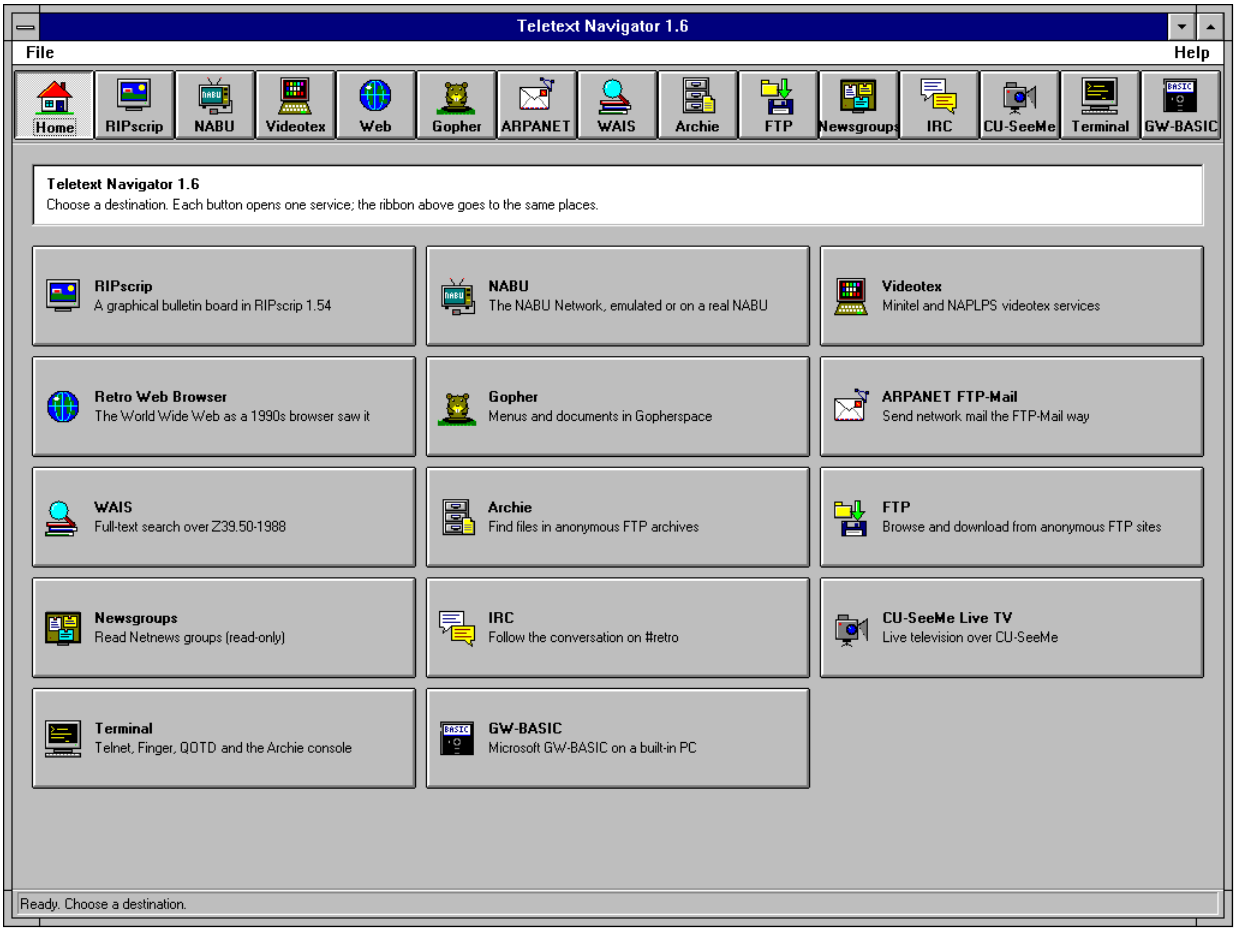



**Figure 2** The Teletext Navigator window, at the Home service.

## Choosing a Service: the Ribbon

Each button on the ribbon opens one service. The button of the service you are using stays pressed in. Point to a button to see the service's full name.

The ribbon is used with the mouse. No key is assigned to a ribbon button.

When you choose another service, the one you leave keeps its place: its screen, its text and, in most services, its connection are still there when you come back. Three services behave differently:

- *CU-SeeMe* disconnects when you leave it, so its sound does not play behind another service. Choose Connect again when you return.
- *NABU* stops its machine or its serial bridge when you leave it.
- GW-BASIC pauses its machine and its sound, and resumes them when you return, with your program still in memory.

The Home (page 11) service, shown when Navigator starts, has one large button for every other service.

## Menu Commands

### File menu

- **Exit** closes Navigator. Any connection that is open is closed.

### Help menu

- **Contents** opens this Help at its Contents.
- **Search for Help on** opens Help's Search dialog box, where you can find a topic by keyword.
- **How to Use Help** explains how to use the Help window.
- **Check for Updates** asks The Teletext whether a newer version of Navigator has been published. See Checking for Updates (page 9).
- **About** shows the version of Navigator, its licence and the credits for the software it contains.

The menus can be opened with the mouse, or with the ALT key and the underlined letter.




**Figure 3** The File menu.




**Figure 4** The Help menu.

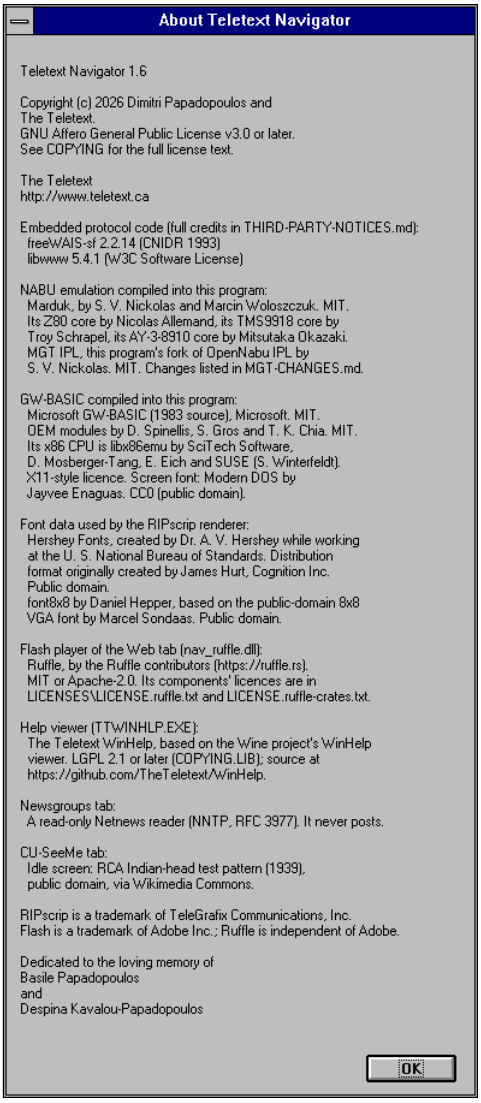



**Figure 5** The About box.

## The Status Bar

The status bar at the bottom of the window shows one line of text from the service you are using: what it is connecting to, what it has received, and any error. It shows "Ready" when you change services.

While a service receives something whose size it knows, a small progress bar at the right end of the status bar shows how much has arrived. The Archie, FTP, Newsgroups, RIPscrip, WAIS and Web services use it.

Some services show messages in their own windows instead: the Terminal prints them in the terminal, Videotex has its own status line under the screen, and CU-SeeMe has a status line under its picture.

## Checking for Updates

Navigator can check whether a newer version has been published.

### Automatic check

About four seconds after Navigator starts, it checks for an update, at most once a day. If the check fails, nothing is shown. To stop the automatic check, clear **Check for updates on startup** in the update dialog box.

### Checking yourself

Choose **Check for Updates** from the Help menu. Navigator reports either that it is up to date, or that a new version is available.

### Installing an update

When a new version is available, a dialog box shows the version and its release notes. Choose:

- **Update Now** to download it. Navigator checks the download against its published SHA-256 checksum, then closes and starts the installer. Open Navigator again when the installer has finished.
- **Remind Me Later** to be asked again at a later start.
- **Skip This Version** not to be offered that version again by the automatic check.

Navigator downloads updates only from apps.teletext.ca. A download that does not match its checksum is deleted and is never run.

## Getting Help

To open Help:

- Choose **Contents** from the Help menu to start at the Contents.
- Press **F1** to open the Help topic for the service you are using.

In three places F1 belongs to the service instead of to Help, because the program or terminal shown there uses F1 itself:

- In GW-BASIC (page 60), F1 is the first function key of the BASIC screen.
- In Videotex (page 19), F1 is the GUIDE key of the Minitel, and the Help key of a NAPLPS service.

In those services, choose Contents from the Help menu. In Videotex you can also click the Host box first, then press F1.

CHAPTER 4



# Home

Home is the service Navigator shows when it starts. It has one large button for each of the other services, in the order of the ribbon, with a line saying what the service is.

To open a service, click its button. Each Home button does the same as the service's button on the ribbon (page 7).

Home has nothing to connect to and no settings.

CHAPTER 5



# RIPscrip

### About RIPscrip

RIPscrip, the Remote Imaging Protocol Script, was developed by TeleGrafix Communications in the early 1990s as a graphical presentation system for information services reached by modem. A text terminal could show letters and ANSI character graphics, but no point-and-click screens, and sending whole pictures over a telephone line was too slow.

RIPscrip solved this by sending short text commands that describe lines, boxes, circles, colours, text, icons and buttons. The terminal draws the screen itself, commonly in the 640 by 350, 16-colour *EGA* format of RIPterm 1.54. Newspapers and public services used it to put wire copy, local news, weather, notices and downloadable documents behind a graphical, mouse-driven screen. Sun.ONE, launched in March 1995 by The Gainesville Sun and the University of Florida College of Journalism and Communications, carried local, national and international news this way. Its time was short: web browsers soon reached everyone. Today active public RIPscrip services are exceptionally rare, and there is no registry from which they can be counted.

The Teletext's RIPscrip service implements the RIPscrip 1.54 command set, reconstructed from TeleGrafix's published specification. Its pictures are drawn at about the speed of a 14,400 bits per second modem; file downloads are not slowed.

### RIPscrip in Navigator

The RIPscrip service is a graphical terminal for *RIPscrip* 1.54 bulletin boards, connected by *Telnet*. You choose the board's buttons with the mouse or with their keys, and receive its files by *Zmodem*. It is set up for The Teletext's board, ripscrip.teletext.ca: news, markets and numbers, weather, traffic, public alerts and the print replica edition.


Connecting to a RIPscrip Board . . . . . 12
RIPscrip Settings . . . . . 13
Using a RIPscrip Board . . . . . 13
RIPscrip Troubleshooting . . . . . 14


## Connecting to a RIPscrip Board

To connect:

1. Check the **Host** box. It starts with ripscrip.teletext.ca.
2. Check the **Port** box. It starts with 23, the *Telnet* port.
3. Choose **Connect**.

The status bar shows "Connecting to" and the host. While the connection is being made the button reads Disconnect and is dimmed; it cannot be stopped, but it gives up after 20 seconds. When the board answers, the status bar shows "Connected to" and the host, the Host and Port boxes are locked, and the screen waits for the board's first picture.

Navigator answers the board's request for its RIPscrip version automatically, as a RIPscrip 1.54 terminal.

To disconnect, choose **Disconnect**. The status bar shows "Disconnected from" and the host, and the screen is cleared. If the board ends the session, the same thing happens. Navigator does not reconnect by itself; choose Connect again.

If you leave the RIPscrip service for another, the session stays open and the screen is kept.

## RIPscrip Settings

The only settings are the **Host** and **Port** boxes, which can be changed while you are disconnected.

The RIPscrip service has no other options, and remembers nothing between sessions: the next time Navigator starts, Host and Port show ripscrip.teletext.ca and 23 again.

## Using a RIPscrip Board

### The screen

The board's 640 by 350 picture is shown at the proportions of a 4:3 monitor, as large as the window allows, with black bars at the sides or the top if needed. While a screen is being drawn the status bar shows how much has been read, and "Page: Done" when it is complete.

### Choosing a button

Click a button on the board's screen. It is shown pressed while you hold the mouse button, and its command is sent when you release it over the same button. To change your mind, move off the button before releasing it.

Many buttons also have a key, usually a letter or a digit shown on the button. Press the key and the button flashes, then its command is sent. Upper and lower case are the same.

When a screen has no buttons, what you type is sent to the board: letters, digits, SPACEBAR, ENTER, BACKSPACE and the hyphen. When a screen has buttons, only their keys work.

### Downloading a file

When you choose a download on the board, or the board starts sending a file, Navigator receives it by *Zmodem*. No dialog box appears:

- The file is saved in your **Downloads** folder.
- A file of the same name is never replaced: the new one is saved as name (1), name (2) and so on.
- The status bar shows the progress, with the progress bar when the size is known.
- Press ESC to cancel. A file that does not arrive completely is deleted.

While a file is being received, other keys and clicks are ignored. When it is done, the screen shows "Saved", the name and the size, and the status bar shows "Transfer: Done".

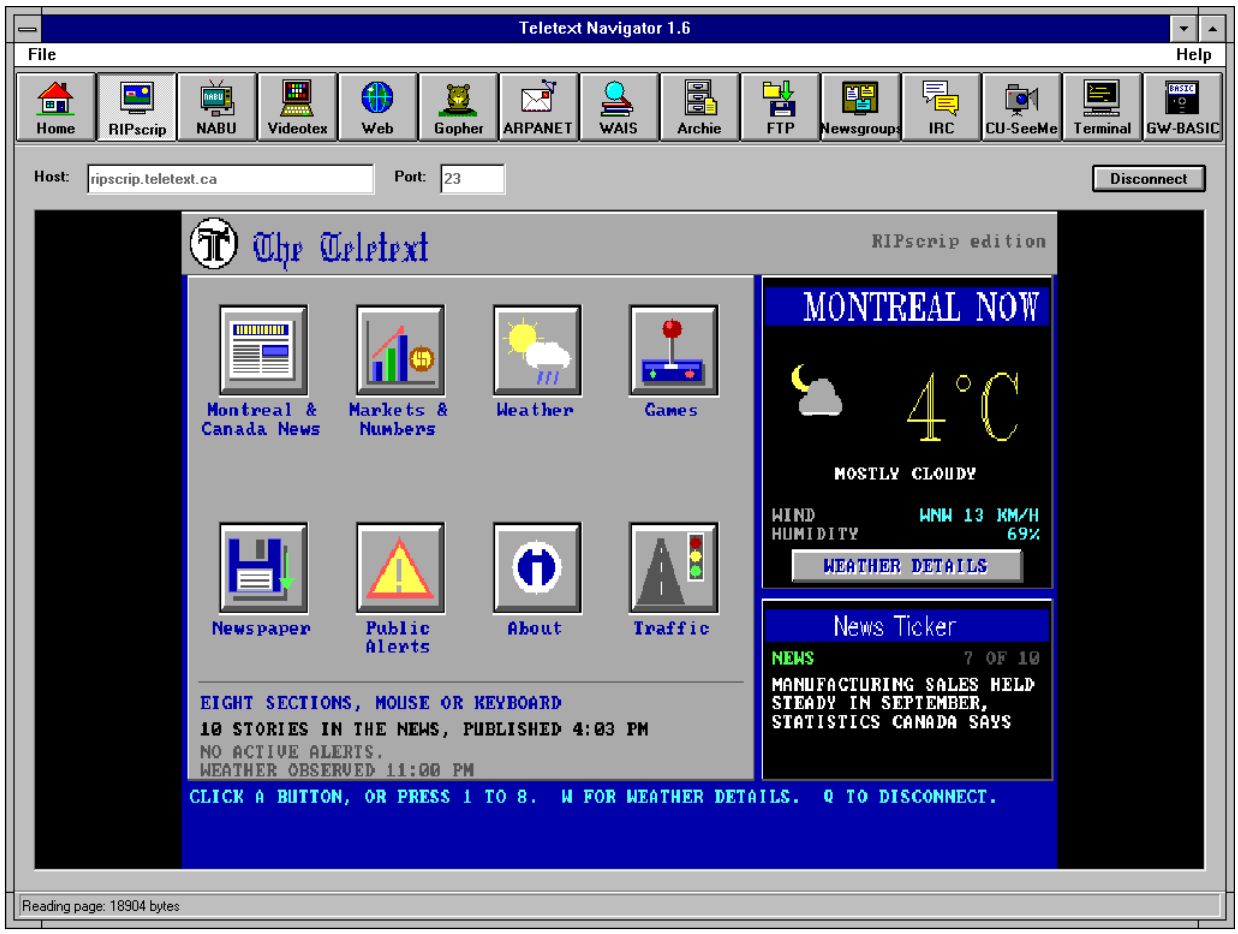



**Figure 6** The RIPscrip service, connected to The Teletext's board at ripscrip.teletext.ca.

## RIPscrip Troubleshooting

### "Enter a host name first."

The Host box is empty. Type a host name and choose Connect.

### "Connection error: Host not found"

The name in the Host box could not be found. Check its spelling; use the host's *DNS name*.

### "Connection error: Connection refused" or "Connection timed out"

The host did not accept a connection on that port. Check the Port box (23 for The Teletext's board), and that your network allows Telnet connections.

### The screen stays black after connecting

The status bar says the session is waiting for the RIPscrip scene. Some hosts send plain text to a terminal that is not RIPscrip; Navigator draws only RIPscrip screens. The Teletext's board at ripscrip.teletext.ca sends RIPscrip.

### Typing does nothing

On a screen with buttons, only the buttons' keys are sent. Click a button instead.

### "This screen has more than 128 clickable regions."

RIPscrip 1.54 allows 128 buttons on one screen. The board sent more; the extra ones do not respond.

### A download stopped

The status bar gives the reason, such as "The connection dropped during the transfer." or "The sender stopped responding." The partial file has been deleted; start the download again from the board.

**"Download held back"**
The receiver could not start. Choose the download again; if it keeps happening, disconnect and connect again.

CHAPTER 6

# NABU

### About the NABU Network

The NABU Network (Natural Access to Bi-directional Utilities) was a Canadian home computer and information service delivered by cable television. Developed in the Ottawa region and launched in 1983, it brought software, news, weather, sports, stock quotations and classified advertising to a computer connected to the household's cable. It has often been called "the Internet before the Internet".

NABU reached about 1,500 subscribers in Ottawa and 700 in Alexandria, Virginia, before the network closed in 1986. A revival began after thousands of unused NABU computers came back on the market in 2022. Only a small number of public NABU services are documented today; The Teletext's NABU edition returns this Canadian invention to active service.

A NABU program was not a page to display: it was sent to the NABU Personal Computer, a Z80 machine, and run there. The honest way to show one is to run one.

### NABU in Navigator

So the NABU service runs a NABU computer: an emulated Z80 machine with its video and sound chips, which starts from its own firmware, MGT IPL, and loads its program from a NABU channel over the network. If you own a real NABU Personal Computer, the service can relay the channel to it through a serial port instead.

The Teletext's NABU channel is channel 1 at nabu.teletext.ca: current news and information as 40-column text and colour graphics.


Starting the NABU . . . . . 16
NABU Settings . . . . . 17
Using the NABU . . . . . 17
NABU Troubleshooting . . . . . 18


## Starting the NABU

### In the emulator

1. In the **Mode** list, choose **Emulator** (the usual choice).
2. Check **Host** (nabu.teletext.ca), **Port** (5816) and **Channel** (1).
3. Choose **Connect**.

The screen shows "Connecting to the Teletext channel", then the NABU starts and loads the channel's program. The status bar shows the host, the channel and the state: Connecting, Booting, then Ready when the program is running.

### On a real NABU Personal Computer

1. Cable the NABU to a serial port of your computer.

2. In the **Mode** list, choose **Real NABU PC (serial)**.
3. Choose the port in the **Serial port** list.
4. Check Host, Port and Channel, and choose **Connect**.
5. Turn on the NABU if it is not already running.

Navigator relays the channel between the network and the NABU at the NABU's speed of 111,860 bits per second. The screen shows a log of the connection under a coloured banner: BRIDGING in green while the relay runs, CONNECTING or RECONNECTING in amber, DISCONNECTED in red. If the network or the serial port drops, Navigator reconnects by itself.

### Stopping

Choose **Disconnect**, or choose another service: leaving the NABU service stops the machine or the relay.

## NABU Settings

- **Mode**: Emulator, or Real NABU PC (serial).
- **Host**, **Port** and **Channel**: where the NABU channel is. The Teletext's is nabu.teletext.ca, port 5816, channel 1. Channels are numbered from 1 to 256.
- **Serial port**: the port the real NABU is cabled to. It can be chosen only in serial mode.

These settings are locked while the NABU is running. Disconnect puts Host, Port and Channel back to The Teletext's channel.

Navigator remembers only the serial port you chose. Mode, host, port and channel start at The Teletext's channel each time.

## Using the NABU

Click the NABU's screen to give it the keyboard. Keys reach the machine only while it is running.

Letters, digits and punctuation are typed as on a US keyboard, whatever keyboard layout Windows uses. SHIFT, CAPS LOCK and CTRL work as on the NABU. The NABU's special keys are on these keys:

- The arrow keys: the NABU's arrow keys.
- PAGE DOWN and PAGE UP: the NABU's right and left keys (the two double-arrow keys).
- DEL: NO. INS: YES.
- ALT: SYM. PAUSE: the NABU's PAUSE key.
- BACKSPACE: the rubout key.

The NABU's TV/NABU key has no key on the PC.

The NABU's sound plays through your computer's speakers. The picture is shown at 4:3, as large as the window allows. The machine always runs at the speed of a real NABU.




**Figure 7** The NABU service: the emulated NABU Personal Computer running the program of The Teletext's channel.

## NABU Troubleshooting

**"No host given.", "Port must be between 1 and 65535." or "Channel must be between 1 and 256."**
Correct the box named, then choose Connect.

**"Could not connect" or "No channel connection"**
The NABU could not reach the channel server, or lost it. Check Host and Port, and your network connection.

**"No NABU firmware found."**
The NABU's start-up firmware is missing from Navigator's folder. Install Navigator again.

**"No serial port chosen."**
In serial mode, choose a port in the Serial port list first.

**The real NABU does not start the program**
Check the cable and the port, and that the banner reads BRIDGING. Some serial adapters cannot run at the NABU's speed of 111,860 bits per second; the log says so if the driver refuses it.

**Keys do nothing**
Click the NABU's screen first. The machine must be running.

CHAPTER 7

# Videotex

### About videotex

Videotex was one of the first attempts to make the telephone line and a television-style screen into an electronic publishing medium. Long before the World Wide Web, a reader could call a service, choose numbered pages from a menu, and read news, weather, sports results, financial information, directories and public information from a central computer. News no longer had to wait for the next printed edition.

In France, newspapers had an early role in the *Minitel* services; Libération used the Minitel to give results of the 1984 Los Angeles Olympic Games through the French night.

Videotex also has a strong Canadian and Montreal history. Canada's Telidon research produced the graphics technology that became *NAPLPS*, which sends text, shapes, colours and fills as compact drawing instructions that the terminal draws itself. It reached the public through Bell Canada's ALEX service, whose commercial trial began in the Montreal area on December 5, 1988. Its Alextel terminal, designed for Bell by Norpak and made by Northern Telecom, reached its services over Bell's DATAPAC network. A Minitel service had entered the Quebec market in the spring of 1988, so the two systems competed in the same city.

### Videotex in Navigator

The Videotex service is a terminal for both kinds of service: the Minitel (40 columns by 25 rows, with its own function keys), which it starts with, and NAPLPS.

The Teletext publishes both editions at videotex.teletext.ca: daily news, markets and numbers, traffic, weather, maps and public alerts. Pages are sent at about 1200 bits per second, some 120 characters a second, so screens build as they did on a period terminal.


Connecting to a Videotex Service . . . . . 19
Videotex Settings . . . . . 20
Using Minitel and NAPLPS . . . . . 20
Videotex Troubleshooting . . . . . 21


## Connecting to a Videotex Service

### Minitel

1. In the **Format** list, choose **Minitel**. The Port box changes to 516.
2. Check the **Host** box (videotex.teletext.ca).
3. Press **CONNEXION/FIN** on the screen keyboard, or F9.

The status line under the screen shows "Connected to" and the host, and a C appears in the top right corner of the Minitel screen (F when it is not connected). To disconnect, press CONNEXION/FIN twice: the first press is sent to the service, the second hangs up.

### NAPLPS

1. In the **Format** list, choose **NAPLPS**. The Port box changes to 1983.
2. Check the **Host** box (videotex.teletext.ca).
3. Choose **Connect**.

To disconnect, choose **Disconnect** or press F10.

### After disconnecting

The screen shows "Disconnected from" and the host, then, five seconds later, the format's opening screen. The format can be changed only while you are disconnected. The session and the screen are kept if you choose another service.

## Videotex Settings

- **Format**: Minitel or NAPLPS. Changing it sets the usual port: 516 for Minitel, 1983 for NAPLPS.
- **Host** and **Port**: where the service is. They are locked while connected.
- **Monochrome**: shows the screen in grey levels, as on a monochrome terminal (for Minitel, the grey levels of the Minitel 1B tube). Clear it for colour.
- **Scanlines** (Minitel only): draws the dark lines between the rows of a television tube.
- **CAPITALS only (Ctrl+C, M)** (Minitel only): see below.

### Capital letters on the Minitel

Usually a letter is sent as you type it: SHIFT or CAPS LOCK gives a capital, as on a PC. Some Minitel services expect capitals, as the Minitel keyboard did. In **CAPITALS only** mode every letter is sent as a capital, and SHIFT gives a lower-case letter, as on the Minitel; CAPS LOCK then does nothing. To switch, check the box, or press CTRL+C and then M, the Minitel's own Fnct C, M command.

None of these settings is remembered after Navigator closes.

## Using Minitel and NAPLPS

### Minitel keys

Click a key on the screen keyboard, or use the PC key:

- ENVOI (send): ENTER or RIGHT ARROW.
- SOMMAIRE (contents): END.
- RETOUR (back): HOME or UP ARROW.
- SUITE (next): TAB or DOWN ARROW.
- ANNULATION (cancel): ESC or LEFT ARROW.
- CORRECTION: BACKSPACE.
- GUIDE: F1. REPETITION: F2. CONNEXION/FIN: F9.

What you type is shown when the service sends it back; the Minitel does not show it by itself. On the Minitel screen CTRL+C is the Fnct C command, not Copy; the Videotex service has no Copy.

### NAPLPS keys

- F1 Help, F2 Refresh, F3 Cancel, F4 Top, F5 Previous, F6 Menu, F7 Next, F8 Send.
- F9 Exit (a key of the PP3 software, not of the ALEXtel terminal).
- F10 disconnects.

Letters, digits, ENTER, BACKSPACE, TAB, ESC and DEL are sent as typed.

### The screen

Both formats are drawn at a whole-number scale: 320 by 250 for the Minitel, 256 by 200 for NAPLPS. The status line counts what has been received and drawn.

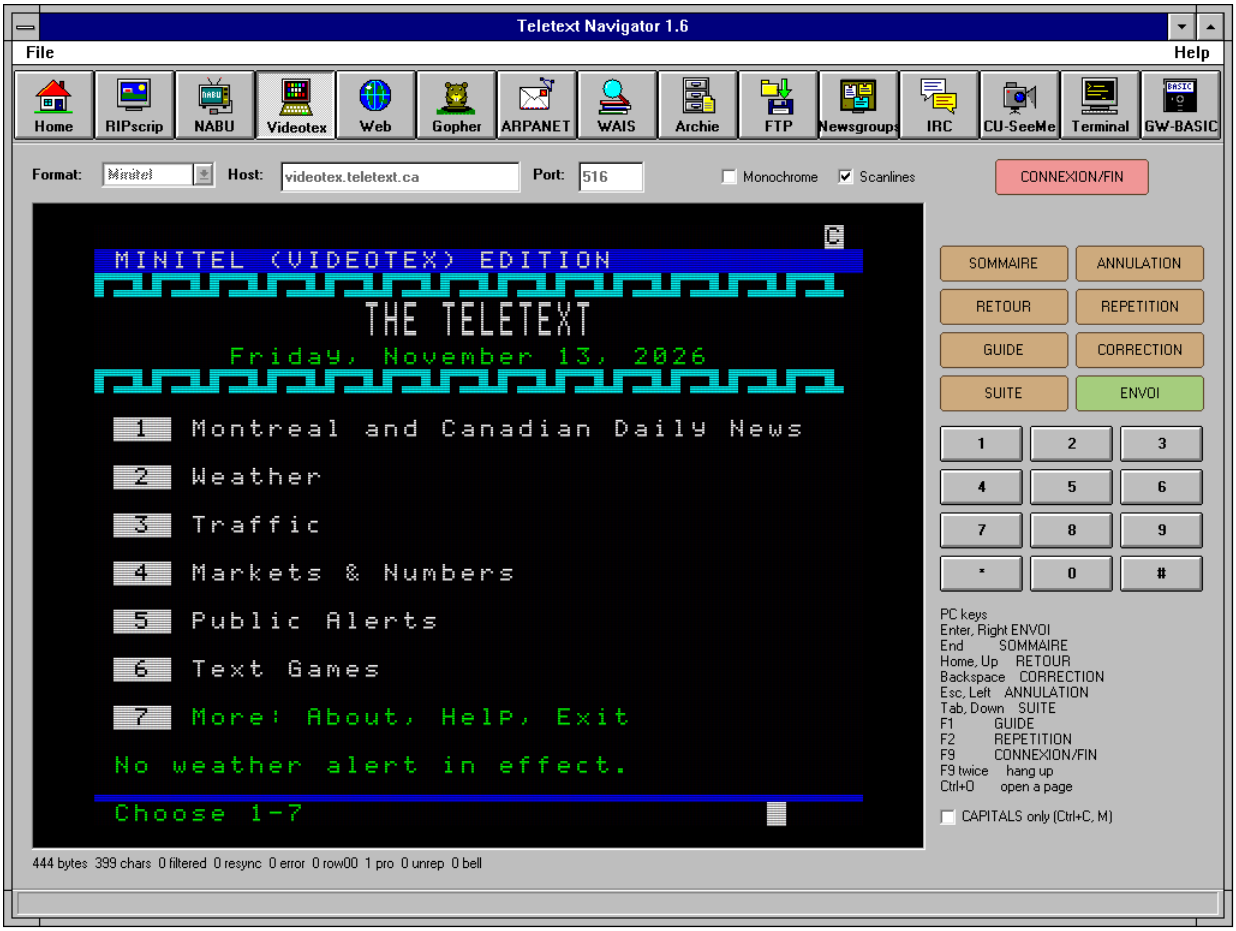



**Figure 8** The Videotex service in Minitel format, connected to videotex.teletext.ca.

## Videotex Troubleshooting

### "Enter a host name" or "Enter a port from 1 to 65535"

Correct the Host or Port box.

### "Connection failed: Host not found"

The host name could not be found. Check its spelling.

### "Connection failed: Connection refused" or "Connection timed out"

Check the port: 516 for The Teletext's Minitel service, 1983 for NAPLPS.

### "Host closed the connection"

The service ended the session. Connect again.

### Nothing appears after connecting

A videotex service speaks first; the screen fills as its page arrives, at about 120 characters a second. Check that the Format matches the service: a Minitel service shown as NAPLPS, or the other way round, does not draw.

### Letters appear in the wrong case

See CAPITALS only in Videotex Settings (page 20).

### Some parts of a NAPLPS picture are missing

Navigator's NAPLPS decoder does not draw macros, redefinable characters, mosaics, unprotected fields or blinking. The status line counts what was skipped.

CHAPTER 8

# Web

### About the World Wide Web of the 1990s

The World Wide Web joined documents on servers all over the Internet with links. Its early pages were written in simple HTML, for browsers such as NCSA Mosaic and Netscape Navigator: headings, paragraphs, lists, tables and pictures, without scripts or style sheets.

In the late 1990s, Internet Explorer 4.0 added Active Channels, a "push" model in which a site's Channel Definition Format (CDF) file listed its contents for the browser to collect, an early ancestor of RSS feeds. To our knowledge, only three Internet Explorer Active Channels remain in operation in the world, The Teletext's among them.

The Teletext also broadcasts Teletext Radio as a RealAudio 3.0 live stream, over the original PNA protocol of mid-1990s Internet radio. To our knowledge, it is the only RealAudio live server using the original PNA protocol in public operation in the world today.

The Teletext's web site, http://www.teletext.ca/, is written for those early browsers: current news, weather, traffic, markets and numbers, Teletext Radio and public alerts, in a classic layout.

### The Web in Navigator

The Web service is a browser of that period. It is built on libwww, the W3C's reference library, and draws pages with its own HTML 3.2 renderer: headings, paragraphs, lists, tables, fonts and colours, and GIF, JPEG and PNG pictures, animated GIFs included. Flash movies (.swf) play through the Ruffle player.


Opening a Web Page . . . . . 22
Web Settings . . . . . 23
Browsing the Web . . . . . 23
Web Troubleshooting . . . . . 26


## Opening a Web Page

The Web service has no connection to open: each page is fetched when you ask for it.

To open a page:

1. Type its address in the **URL** box, for example http://www.teletext.ca/.
2. Press ENTER or choose **Go**.

The status bar shows "Connecting to" and the host, then how much of the page has arrived, then "Document: Done".

- **Home** opens http://www.teletext.ca/.
- **Refresh** fetches the current page again.
- **Back** and **Forward** move through the pages you have seen.

While a page is being fetched, the buttons and the URL box are dimmed except Stop. **Stop** discards the page when it arrives and shows "Stopped."; the transfer itself runs to its end.

Addresses beginning gopher:// open in the Gopher (page 28) service, and addresses beginning ftp:// in the FTP (page 42) service.

## Web Settings

The Web service has no settings. It remembers the last 64 pages for Back and Forward while Navigator runs, and nothing after it closes.

## Browsing the Web

### Links

Point to a link and the pointer becomes a hand, and the status bar shows its address. Click to follow it.

### Scrolling

Use the scroll bar, the mouse wheel, or UP ARROW, DOWN ARROW, PAGE UP, PAGE DOWN, HOME and END.

### Copying text

Drag across text to select it, then press CTRL+C, or click with the right mouse button and choose Copy. CTRL+A selects the whole page; ESC clears the selection.

### Sound

- A link to an .au file plays it.
- A link to a .tsp file plays its TrueSpeech sound.
- A link to a .ram file is passed to the program Windows uses for RealAudio files, if one is installed.

### Flash movies

A movie in a page plays in its place; an address ending in .swf fills the window. Click a movie to give it the mouse and keyboard. When you leave the Web service the movie stops; it starts again from its first frame when you return.

### Active Channels

An address of a .cdf Active Channel file is read, and the status bar reports how many items it has. The channel itself is not displayed.

The Web service cannot save pages or files to disk.




**Figure 9** The Web service at http://www.teletext.ca/.




**Figure 10** The Teletext's Flash animation, a Flash version 4 movie, playing in a page of the Web service (http://www.teletext.ca/flash/demo.htm).




**Figure 11** The Radio page of The Teletext's web site, with its RealAudio link.

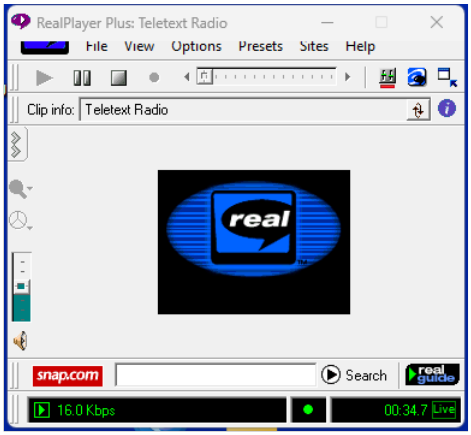



**Figure 12** Teletext Radio playing in RealPlayer, opened from the RealAudio link (its channel sidebar closed).

## Web Troubleshooting

### "Could not load" and the address

The page could not be fetched. Check the address, and that the server is reachable.

### A page looks wrong or empty

Like the browsers of its time, the Web service does not run JavaScript, apply style sheets, use cookies or show frames, and it cannot fill in forms. Pages written for modern browsers may not work. The Teletext's site at http://www.teletext.ca/ is written for it.

### Secure (https://) pages do not open

The Web service fetches pages by plain HTTP, as browsers did before secure pages were common. It cannot open https:// addresses.

### A picture shows as [image: ...]

The picture could not be loaded or read.

### "Flash movie cannot play"

The status bar gives the reason, such as "not a Flash movie", "the movie is larger than 32 MB" or "the movie could not be downloaded".

CHAPTER 9



# Gopher

### About Gopher

*Gopher* is a predecessor of the World Wide Web, created at the University of Minnesota in 1991. It shows menus: each item leads to another menu, a document, a search or a file. There are no scripts, no advertising and nothing for the reader's computer to run.

Before the Web took over, Gopher was one of the Internet's first public news racks. Readers browsed online magazines, political commentary, institutional bulletins, government documents and newspaper-like services through simple menus. In public debates such as the one over the North American Free Trade Agreement (NAFTA), Gopher led readers to current commentary and the related documents, while online news publishing was still an experiment. Universities, media services, libraries and public institutions ran Gopher servers as a searchable newsstand and document shelf.

The Teletext uses Gopher in the same tradition, as a menu-based news rack. Today roughly 200 Gopher servers are estimated to remain active in the world, and The Teletext operates one of them as part of a live publishing system.

### Gopher in Navigator

The Gopher service is a Gopher client (RFC 1436). It opens menus and text documents, shows pictures, runs Gopher searches, plays sounds and saves files. It is set up for The Teletext's Gopher server, gopher.teletext.ca: daily news, weather, traffic, markets and numbers, public alerts, articles, archives and the print replica edition in PDF.


Opening a Gopher Menu . . . . . 28
Gopher Settings . . . . . 29
Using Gopher Menus . . . . . 29
Gopher Troubleshooting . . . . . 30


## Opening a Gopher Menu

Gopher has no connection to keep open: each menu or document is fetched when you ask for it.

To open a menu:

1. Type its address in the **Selector** box, for example gopher://gopher.teletext.ca/. You may leave out gopher://, and add a port after the host, as in gopher.teletext.ca:70.
2. Press ENTER or choose **Go**.

The status bar shows "Connecting...", then how many bytes have arrived, then "Menu: Done" or "Document: Done".

- **Home** opens gopher://gopher.teletext.ca/.

- **Refresh** fetches the current menu or document again.
- **Back** and **Forward** move through what you have seen.

Nothing is fetched until you choose Go or Home.

## Gopher Settings

The Gopher service has no settings. It remembers the last 64 menus and documents for Back and Forward while Navigator runs, and nothing after it closes. The usual Gopher *port*, 70, is used unless the address names another.

## Using Gopher Menus

Each menu item has an icon for its kind. Items you can open are shown in blue; point to one and the pointer becomes a hand. Click to open it:

- **Menus** and **text documents** open in the window.
- **Pictures** (GIF and other images) open in a window of their own. Close it with its Control-menu box.
- **Searches** open the Gopher Search dialog box. Type the words to find and choose OK; the results are shown as a menu.
- **Sounds** play through your speakers.
- **Files** open a Save As dialog box, so you can choose where to save them. Files whose names end in .mp3 or .au are played instead.

Telnet, video and archive items are shown with their icons, but cannot be opened.

### Copying text

Drag across text to select it (a click without dragging opens the item under the pointer). Press CTRL+C to copy, CTRL+A to select everything, or click with the right mouse button for the same commands.

### Scrolling

Use the scroll bar, the mouse wheel, or the arrow keys, PAGE UP, PAGE DOWN, HOME and END.

Gopher links followed in the Web (page 22) service open here.

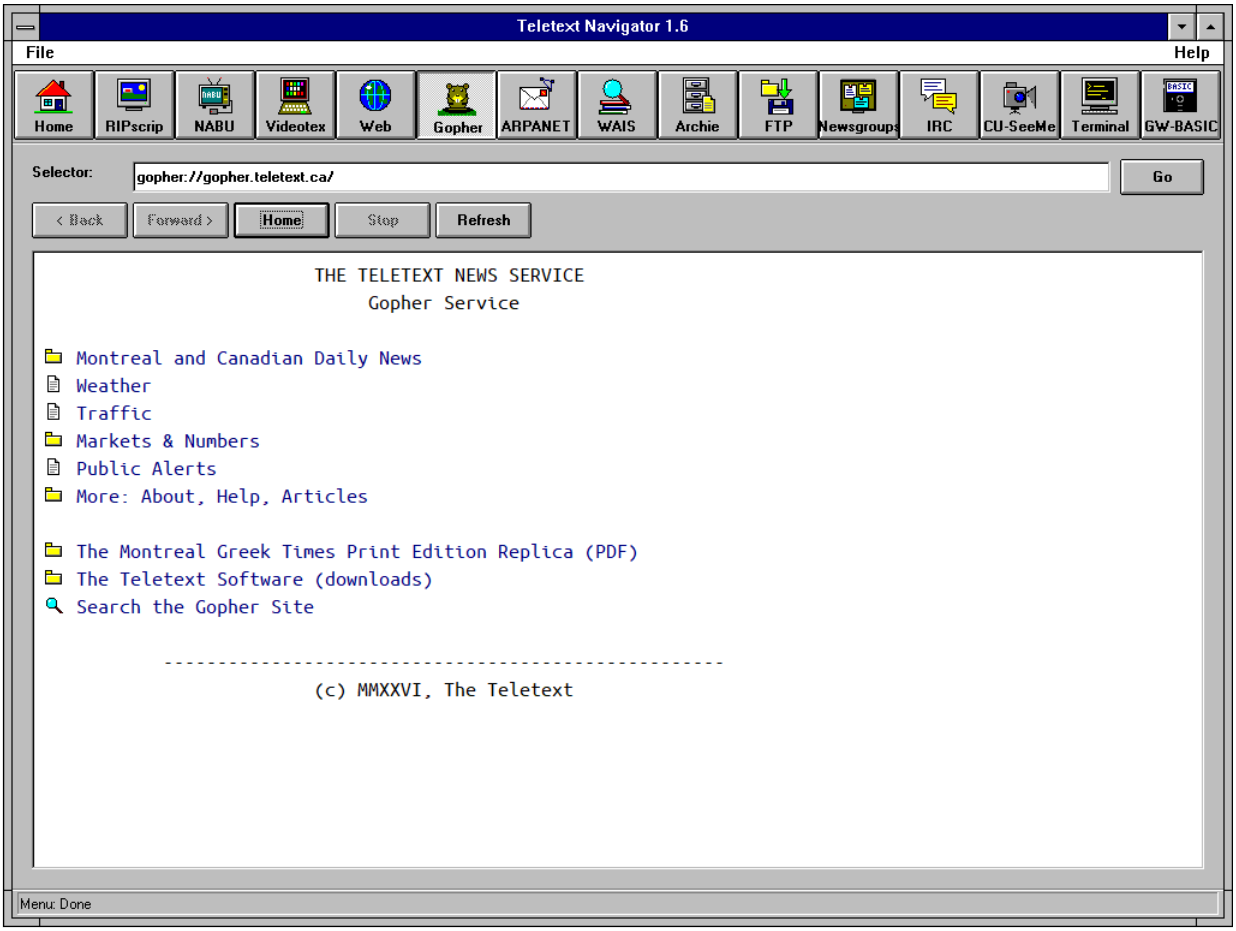



**Figure 13** The Gopher service at gopher.teletext.ca.

## Gopher Troubleshooting

### Stop does not stop at once

Stop asks the transfer to end; the Gopher service waits until the server stops sending, at most 15 seconds.

### "a fetch is already in flight; press Stop first."

Only one menu or document is fetched at a time. Wait for it, or choose Stop.

### "malformed URL."

The address in the Selector box is not a Gopher address. It should look like gopher://host/ or gopher://host/1/path.

### An item does nothing when clicked

Some kinds of item (Telnet sessions, video, archives and Web links) cannot be opened by the Gopher service.

### "image decode failed."

The picture could not be read.

### "Audio service busy (Stop first)."

A sound is already playing.

CHAPTER 10



# ARPANET

### About the ARPANET and FTP-Mail

Before today's Internet, there was the ARPANET, a network created in the United States to connect universities, research centres, government agencies and laboratories. On January 1, 1983, it moved from its older Network Control Protocol (NCP) to TCP/IP, which laid the foundation of the Internet we know.

At first, electronic mail could not travel from one computer to another. In 1971 Ray Tomlinson made it possible on the ARPANET for a message to move between connected computers. That idea became part of the history of *FTP-Mail*, in which mail was sent with commands of the File Transfer Protocol.

The Teletext has recreated FTP-Mail as a working service, with the command flow of the old Internet specifications. Messages sent to it are received by The Teletext's mail system and may be read by the operator or the editors.

### ARPANET in Navigator

The ARPANET service sends a short message by FTP-Mail, with the MAIL command of RFC 765. No account or password is needed. It is set up for The Teletext's server, arpanet.teletext.ca, port 2121, and its mailbox INFO@TELETEXT.


Connecting to an FTP-Mail Server . . . . . 31
ARPANET Settings . . . . . 31
Writing and Sending a Message . . . . . 32
ARPANET Troubleshooting . . . . . 33


## Connecting to an FTP-Mail Server

1. Check the **Server** box (arpanet.teletext.ca) and the **Port** box (2121).
2. Choose **Connect**.

Navigator opens a connection, waits for the server's greeting and closes the connection again. The status bar shows each line the server sends. When the greeting arrives, the To and Message boxes and the Send and Clear buttons become available, and Server and Port are locked.

FTP-Mail keeps no connection open between messages: each Send opens its own, and so does Disconnect.

To disconnect, choose **Disconnect**. Navigator says goodbye to the server, clears the message and puts the To box back to INFO@TELETEXT.

## ARPANET Settings

- **Server** and **Port**: the FTP-Mail server. The Teletext's is arpanet.teletext.ca, port 2121.

- **To**: the mailbox. It starts as INFO@TELETEXT. A mailbox may be written as NAME@HOST or NAME AT HOST.

The ARPANET service remembers nothing after Navigator closes.

## Writing and Sending a Message

1. Connect, as described in Connecting to an FTP-Mail Server (page 31).
2. Check the **To** box, and press ENTER to move to the message.
3. Type the message in the **Message** box. ENTER starts a new line.
4. Choose **Send**, or press CTRL+ENTER.

Under the message, **Bytes used** counts its size against the server's limit of 2048 bytes, and says "over cap" when it is too long.

Navigator sends the message with the MAIL command and shows the server's replies in the status bar. When the server accepts it, the message is locked and Send is dimmed, so it is not sent twice. Choose **Clear** to empty the message and write another; the To box is kept.

Before sending, line endings are made standard, and control characters (except TAB) and terminal escape sequences are removed.

A line holding only a period ends a message in FTP-Mail. Pressing ENTER at the end of such a line sends the message, and a line of only a period cannot be part of a message: everything after it is not sent.

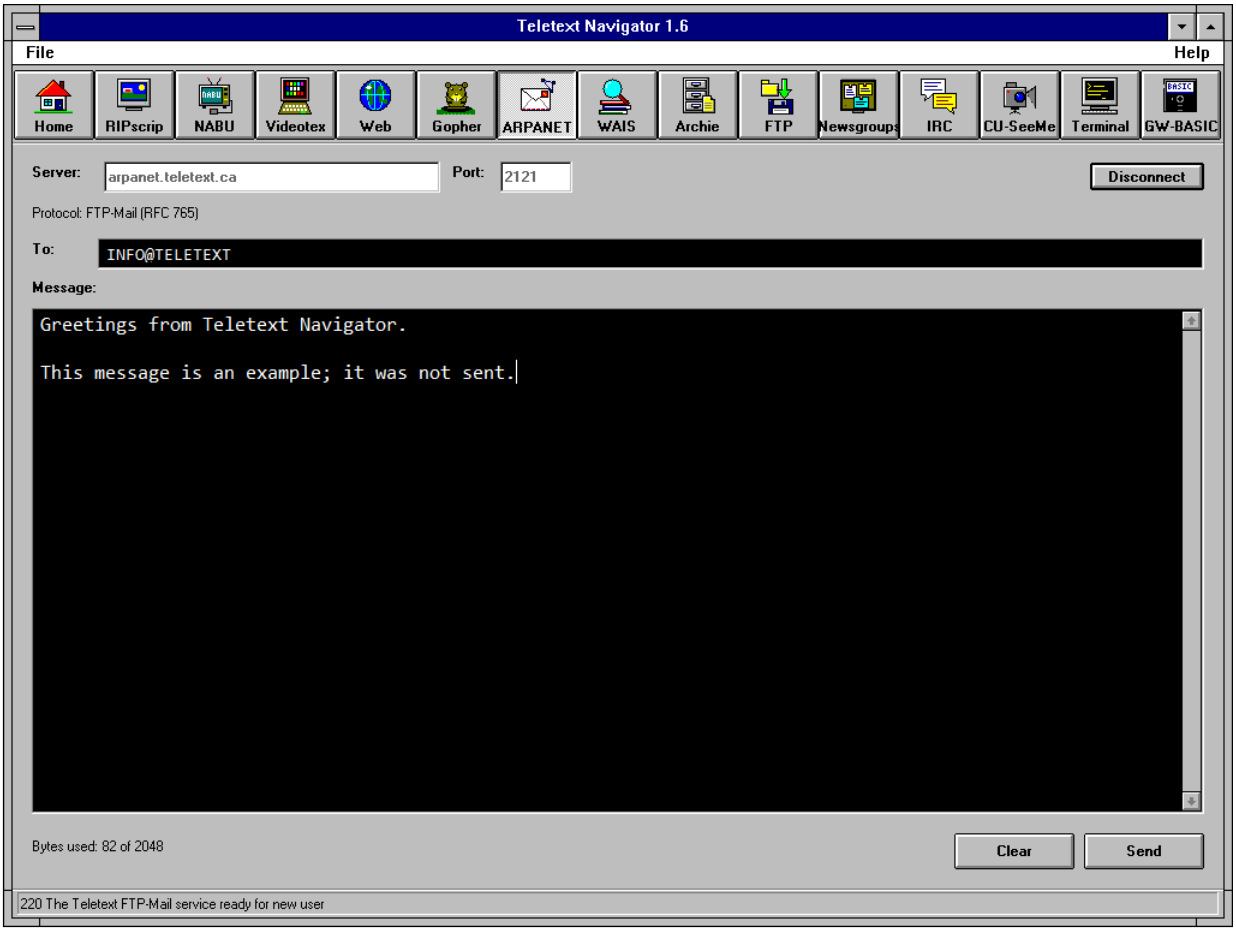



**Figure 14** A message written in the ARPANET service, ready to send (this one was not sent).

## ARPANET Troubleshooting

**"Server and Port are required." or "Port must be 1-65535."**
Correct the box, then choose Connect.

**"Cannot resolve" and the server name**
The server's name could not be found. Check its spelling.

**"Cannot connect to" the server**
The server did not answer on that port. The Teletext's FTP-Mail server uses port 2121.

**"No FTP-Mail banner from" the server**
The server answered but did not greet as an FTP-Mail server. Check the port.

**The server's reply begins with 550**
The mailbox in the To box is not known to the server.

**The server's reply begins with 552**
The message is longer than the server accepts. Shorten it to 2048 bytes or less.

CHAPTER 11



# WAIS

### About WAIS

*WAIS*, the Wide Area Information Server, was one of the first distributed search systems on the Internet. Instead of browsing menus, you asked a question: you send words to a WAIS server, and it returns the documents of a database ranked by how well they match. It came out of early Internet research into information discovery.

WAIS carried its searches in *Z39.50-1988*, the American national standard for information retrieval of its day.

The Teletext runs a live WAIS server with the original WAIS protocol. To our knowledge, it is the only publicly accessible production WAIS server using the original WAIS protocol, derived from Z39.50-1988, in operation in the world today. It is not a preserved relic but a working search service attached to a news operation: a ranked full-text search of The Teletext's articles.

### WAIS in Navigator

The WAIS service is a native WAIS client. It speaks Z39.50-1988 (Version 1), the protocol of the original WAIS servers, with client code from freeWAIS-sf 2.2.14. It is set up for The Teletext's server, wais.teletext.ca, port 210, database teletext.


Connecting to a WAIS Server . . . . . 34
WAIS Settings . . . . . 35
Searching and Reading Documents . . . . . 35
WAIS Troubleshooting . . . . . 37


## Connecting to a WAIS Server

1. Check the **Server** box (wais.teletext.ca), the **Port** box (210) and the **Database** box (teletext).
2. Choose **Connect**.

Navigator checks that the server can be searched:

- It opens a connection to the server, within 3 seconds.
- It sends the Z39.50-1988 initialization, and waits 3 seconds for the answer.
- It runs a test search in the database you named.

While it checks, the status bar shows "Connecting..." and the button reads Disconnect but is dimmed; the check cannot be stopped. When the test search succeeds, the status bar shows "Connected to" and the server and database. The Search box, the Doc # box and their buttons become available, and Server, Port and Database are locked.

WAIS keeps no connection open: each search and each document you read opens a connection of its own. "Connected" means that the server and database answered the check.

To disconnect, choose **Disconnect**. Nothing is sent to the server; Navigator stops any search in progress and clears the Search box, the Doc # box and the results.

## WAIS Settings

- **Server**: the WAIS server's name. The Teletext's is wais.teletext.ca.
- **Port**: 210, the usual WAIS port.
- **Database**: the database to search on that server. The Teletext's is teletext.

These can be changed only while you are disconnected. They are kept when you choose another service, but not after Navigator closes.

The line **Protocol: Z39.50-1988** names the protocol; it cannot be changed.

## Searching and Reading Documents

### Searching

1. Type one or more words in the **Search** box.
2. Press ENTER or choose **Search**.

The status bar shows "Searching...", then "Search: Done" and the number of documents. The server returns at most 40.

The results begin with "Search Response:" and the number of records returned. Each document then has one line: its number, a bar of up to nine solid blocks showing its score (the server's own measure of how well it matches; a longer bar is a better match), and its headline, or "(no title)" if it has none.

### Reading a document

1. Type the document's number in the **Doc #** box.
2. Press ENTER or choose **View**.

The status bar shows how much of the document has arrived, then "Document: Done". The document replaces the results in the window. A document is read up to its full length, at most one megabyte.

Choose < to go back from the document to the results, and > to go forward to the last document again.

### Moving and copying

- UP ARROW, DOWN ARROW, PAGE UP and PAGE DOWN scroll the window even while you are typing in a box.
- ESC stops a search or a document that is arriving.
- To copy, select text in the window and press CTRL+C, or click with the right mouse button and choose Copy. CTRL+A selects everything.

Only one search or document can be in progress at a time.

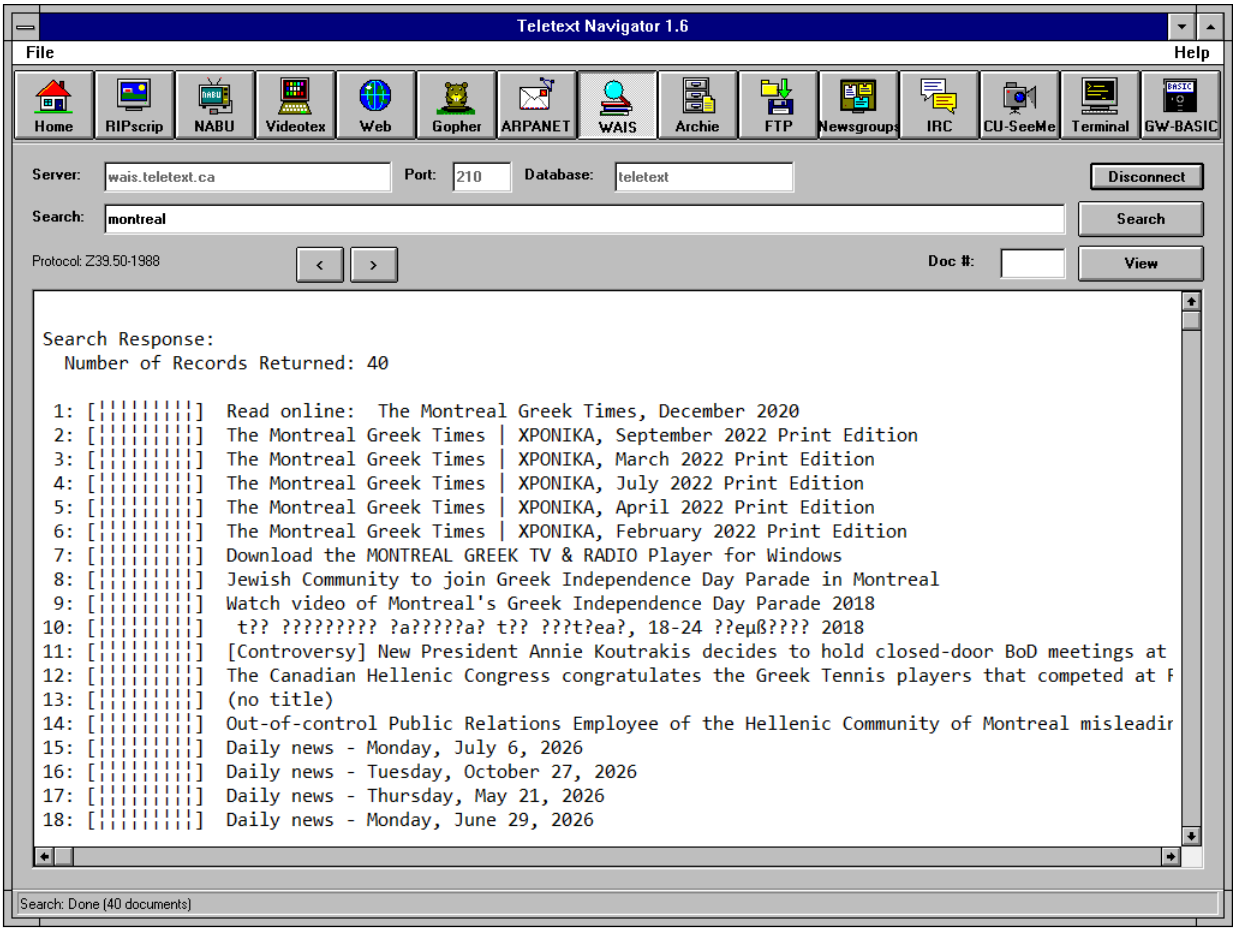



**Figure 15** A WAIS search of The Teletext's database for the word "montreal".

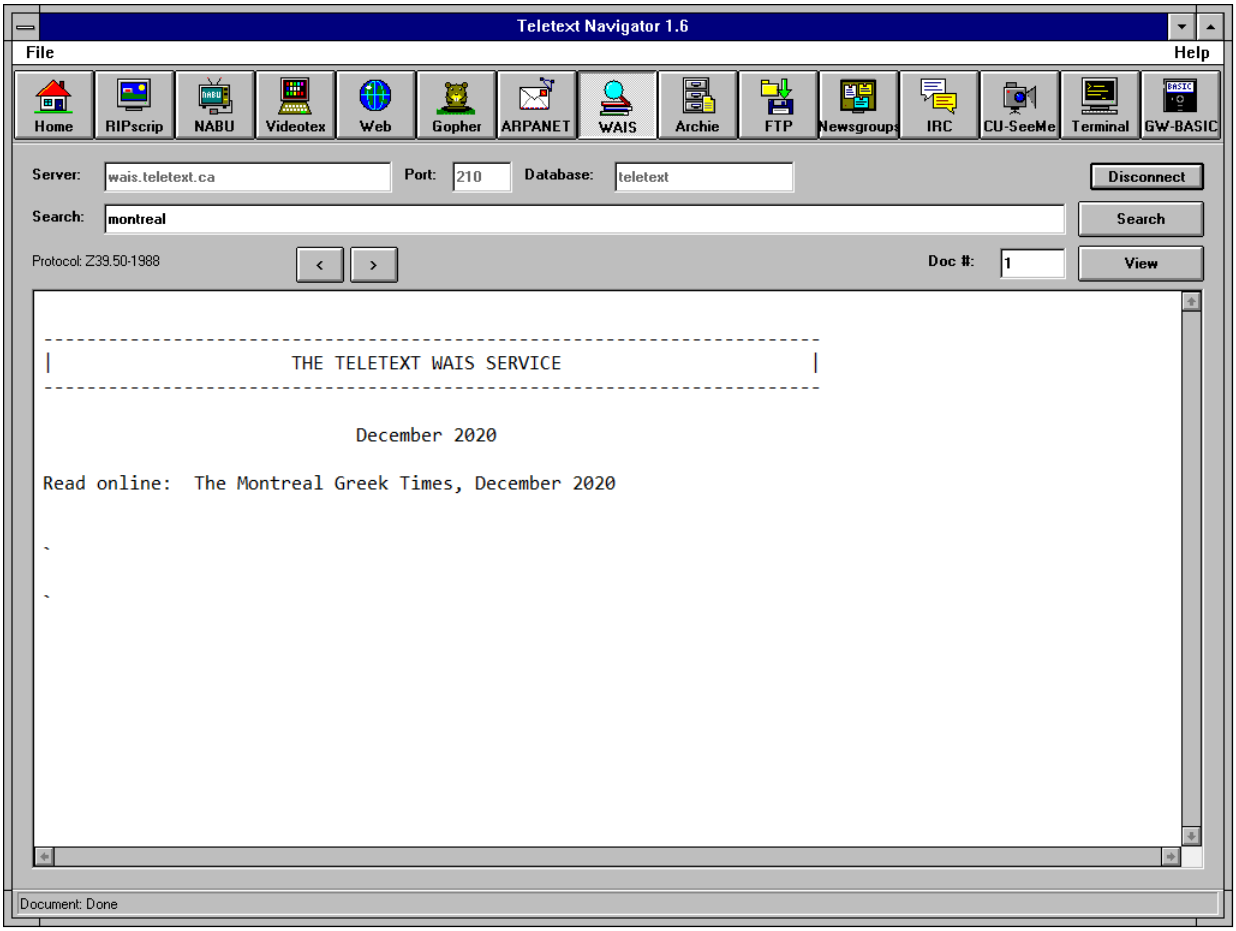



**Figure 16** A document read from the WAIS server.

## WAIS Troubleshooting

**"Server, port, and database are required." or "Port must be 1-65535."**
Fill in the box, then choose Connect.

**"Cannot connect to" the server**
The server did not accept a connection on that port within 3 seconds. Check the name and the port (210).

**"does not speak WAIS Z39.50-1988."**
Something answered on that port, but not a WAIS server.

**"Database" ... "not found"**
The server does not have a database of that name. The Teletext's database is teletext, in lower case.

**"Run a search first." or "Enter a Doc # from 1 to" a number**
Search before reading, and use a number from the results.

**"WAIS is busy: press Esc to cancel."**
A search or a document is still arriving. Wait, or press ESC.

**A search or document never finishes**
Press ESC to stop it, then try again. Navigator waits as long as the server takes.

**"No response from server." or "Z39.50-1988 Init failed."**
The server closed the connection or did not answer the search. Try again; if it continues, disconnect and connect again.

CHAPTER 12



# Archie

### About Archie

*Archie* was the Internet's first search engine, created at McGill University in Montreal in 1989 by Alan Emtage, with Peter Deutsch and Bill Heelan. Before the World Wide Web, public anonymous FTP servers held a large and growing collection of software, documents, reports, data, pictures, sound and public records. Finding them was the problem. Archie solved it by collecting the file listings of FTP servers into one searchable index of file names, directories and hosts.

Reporters, librarians and researchers used Archie to find source documents, public records and files connected with the news. During the debate over the North American Free Trade Agreement, the NAFTA text and related documents were distributed through university and legal FTP archives, and library guidance of the time records using Archie to find a copy. Archie was the early Internet's finding desk: it told you which server held a file and where.

Archie was invented here in Montreal, and its return has a special meaning for us. The Teletext runs a Bunyip Archie 3.5 server connected to its own anonymous FTP archive, reached through Archie's original protocol, Prospero's *ARDP*. To our knowledge, it is the only publicly accessible Archie server in operation in the world today. Welcome home, Archie!

Archie's creators were Alan Emtage, Bill Heelan and Peter Deutsch (McGill University, 1989). The Archie 3.5 release came from Bunyip Information Systems, Inc. (1996). Its last surviving copy was preserved by ICM, University of Warsaw, and found and set free by The Serial Port (2024).

### Archie in Navigator

The Archie service asks an Archie server which anonymous FTP sites hold a file, shows the hosts, directories and files it finds, and downloads the one you choose. It speaks ARDP over UDP, as the original clients did. It is set up for The Teletext's server, archie.teletext.ca, which indexes the files of ftp.teletext.ca.


Searching an Archie Server . . . . . 39
Archie Settings . . . . . 40
Choosing and Downloading a File . . . . . 40
Archie Troubleshooting . . . . . 41


## Searching an Archie Server

Archie has no Connect button: each search is one exchange with the server.

1. Type part or all of a file name in the **Search** box, for example pdf.
2. Check the **Archie Server** box (archie.teletext.ca) and the **Port** box (1525).
3. Choose a **Search Type**.
4. Press ENTER or choose **Search**.

The status bar shows "Searching Archie...", and, when the answer comes in several packets, which packet has arrived. When it is complete it shows "Search: Done" and the number of hits. A server returns at most 100 hits.

Navigator tries four times, waiting four seconds each time, before it reports that the server does not answer. A search cannot be stopped once it has started.

## Archie Settings

### Search Type

- **Substring**: finds names that contain the text, in upper or lower case.
- **Substring (case sensitive)**: the same, with upper and lower case kept apart.
- **Exact**: finds only names that are exactly the text.
- **Regular Expression**: the text is a pattern, as in the UNIX ed editor.

**Exact first**: look for an exact match before the substring or pattern search. It is dimmed when Exact is chosen.

### Server

**Archie Server** and **Port** name the server. The Teletext's is archie.teletext.ca, port 1525. You may also type the port after the name, as archie.teletext.ca:1525. Archie uses UDP; the Protocol line says so and cannot be changed.

These settings are kept when you choose another service, but not after Navigator closes.

## Choosing and Downloading a File

The hits are shown in three lists:

- **Hosts**: the FTP sites that hold a match.
- **Directories**: the directories on the chosen host.
- **Files**: the files in the chosen directory.

After a search, the first host and its first directory are chosen for you. Click a host or a directory to fill the next list.

Click a file to see its details under **Selected file**: its name, its size in kilobytes, its UNIX permissions (Mode), its date, and the address of its host.

### Downloading the file

1. Choose a file in the Files list.
2. Choose **Download**.
3. In the Download file dialog box, choose where to save it, and choose Save.

Navigator downloads the file from its host by *anonymous FTP*, in *binary mode* and *passive mode*, on port 21. The status bar counts the bytes, and shows "Saved" when the file is complete. While it downloads, the Download button reads **Cancel**; choose it to stop. A file that does not arrive completely is deleted.

To browse the host's other files, use the FTP (page 42) service.

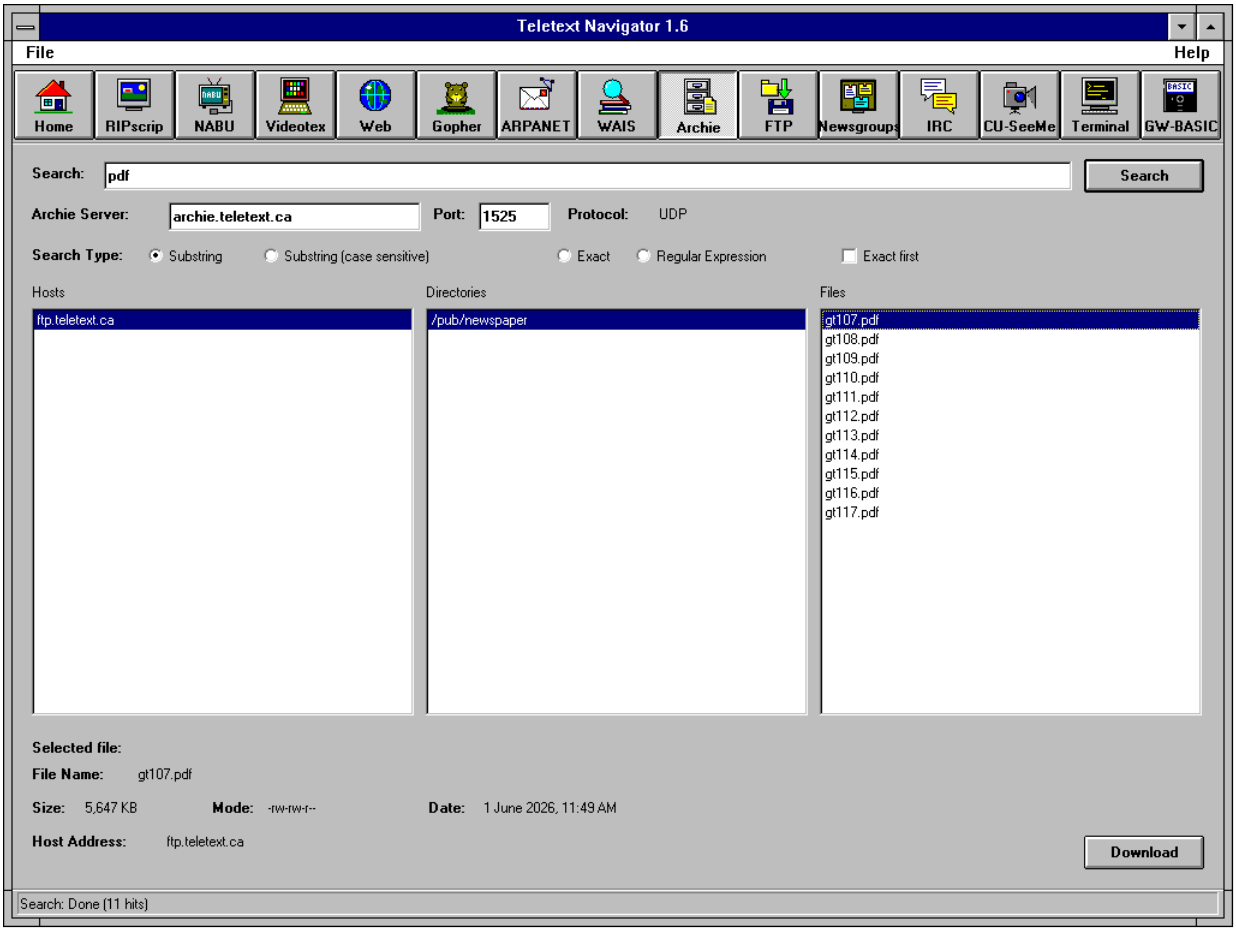



**Figure 17** The Archie service: the hosts, directories and files found for "pdf".

## Archie Troubleshooting

### "Enter a search term and press Search."

The Search box is empty.

### "Cannot resolve" the server

The Archie server's name could not be found. Check its spelling.

### "No response from" the server

The server did not answer after four tries. Archie uses UDP port 1525; some networks block UDP.

### "Search: Done (0 hits)"

No file name matched. Try Substring instead of Exact, or a shorter part of the name.

### The server's own message, beginning ERROR or FAILURE

The server refused the search; its message says why. A Regular Expression search with a mistake in the pattern is a common cause.

### "Anonymous login failed" or "Cannot connect to" a host

The FTP site that holds the file refused the download, or is no longer there. Archie's index may be older than the site.

### "Server refused passive mode."

The host does not allow passive transfers. Archie downloads always use passive mode; try the FTP service with Passive cleared.

CHAPTER 13



# FTP

### About FTP

*FTP*, the File Transfer Protocol, is one of the foundation services of the early Internet, and still one of the simplest ways to offer files to every kind of computer. In the news business, *anonymous FTP* became the Internet's first file counter: the place to fetch the documents, reports, data, pictures, sound, software and public records behind the day's news.

During the NAFTA debate of the early 1990s, anonymous FTP let readers fetch the treaty's text, chapters and appendixes straight from Internet archives. During the war in Bosnia in the mid-1990s, FTP sites gathered and distributed news, documents, pictures and reports about the conflict. FTP was not the headline ticker: it was the document room behind the newsroom.

The Teletext uses FTP in the same way, as the file depot of its system.

### FTP in Navigator

The FTP service lists the files on an anonymous FTP server and copies them to your computer. It is laid out like the FTP programs of the 1990s: your computer's folders on the left, the server's directories on the right, and below them a log of everything said between Navigator and the server.

It downloads only. It logs in as anonymous, and cannot upload, delete, rename or create anything on a server. It is set up for The Teletext's server, ftp.teletext.ca: print replica editions in PDF, audio and video podcasts, software and other files.


Connecting to an FTP Server . . . . . 42
FTP Settings . . . . . 43
Browsing and Downloading Files . . . . . 43
FTP Troubleshooting . . . . . 44


## Connecting to an FTP Server

1. Check the **Host** box (ftp.teletext.ca) and the **Port** box (21).
2. Choose **Connect**, or press ENTER in the Host box.

The status bar shows "Connecting to" and the host. Navigator then:

- opens the connection, within 20 seconds;
- waits for the server's greeting;
- logs in as anonymous, giving navigator@teletext.ca as the password if the server asks for one;
- asks for the current directory and lists it.

The remote list fills, and the status bar shows "Connected to" the host, the directory and the number of entries. Every command and reply appears in the log.

The **Username** box always shows Anonymous; it cannot be changed. A server that does not accept anonymous logins cannot be used.

To stop a connection that is being made, choose the button again; it reads Disconnect while connecting. To disconnect, choose **Disconnect**: Navigator sends QUIT and shows the server's goodbye.

### FTP addresses

An ftp:// address followed in the Web (page 22) service opens here: Navigator connects to the host, goes to the directory, and selects the file if the address names one. A user name in the address is ignored.

### Idle sessions

Navigator does not keep an idle session alive. Many servers end a session that has been idle for some minutes; the status bar then says so, and you connect again.

## FTP Settings

### Passive

When **Passive** is checked (as it is at first), your computer opens the data connection for each listing and file. This is *passive mode*, and it works through most firewalls and home routers. Clear it only if a server needs the older active mode, in which the server connects to your computer.

### Transfer

- **ASCII**: every file is sent in *ASCII mode*, as text.
- **Binary**: every file is sent in *binary mode*, unchanged.
- **Auto** (chosen at first): text files are sent in ASCII mode and everything else in binary mode. Auto treats as text the names ending .txt, .asc, .me, .1st, .diz, .nfo, .htm, .html, .md, .ini, .bat, .c, .h, .sha256, .md5, .sfv, .csv, .log, .xml, .json, .c and .lst, and names without an extension that begin README, INDEX or LICENSE, or are 00INDEX.

### Host and Port

Change them while you are disconnected. The usual FTP port is 21.

None of these settings is remembered after Navigator closes.

## Browsing and Downloading Files

### The server's files (right)

Directories and links are listed first, then files, each in alphabetical order. Each line shows the name, the size in bytes and the date. ".." goes up one level.

- Double-click a directory, or select it and press ENTER, to open it.
- Double-click a file, or select it and press ENTER, to download it.
- A link that leads to a directory opens it; otherwise the link is downloaded.

### Your folders (left)

The local side starts in your Downloads folder. Double-click a folder, a drive or ".." to move. Downloads are saved in the folder shown on the left.

### Downloading

Select a file on the right and choose <--, or double-click it. No dialog box appears:

- The file is saved in the local folder on the left.
- A file of the same name is never replaced: the new one is saved as name (2), name (3) and so on.
- Characters that Windows does not allow in names are replaced with _.
- The status bar shows the progress, with the progress bar when the server gives the size.

When the file is complete, the status bar shows "Transfer: Done", the log shows its size and time, and the local list is refreshed.

To cancel a download, press ESC in either list or in the log. The session stays open. A file that does not arrive completely is deleted.

### Viewing a file

Select a file and choose **View** to open it in the program Windows uses for that kind of file. View opens only text files (.txt, .sha256, .asc, .log, .nfo, and files named 00INDEX, INDEX, README or CHANGES) and PDF files; it never opens a program or a script. A file on the server is first copied to a temporary folder, which is removed when you disconnect.

### The log

Lines from the server are shown in black, the commands Navigator sends in blue, and Navigator's own notes in grey. The last 3000 lines are kept.

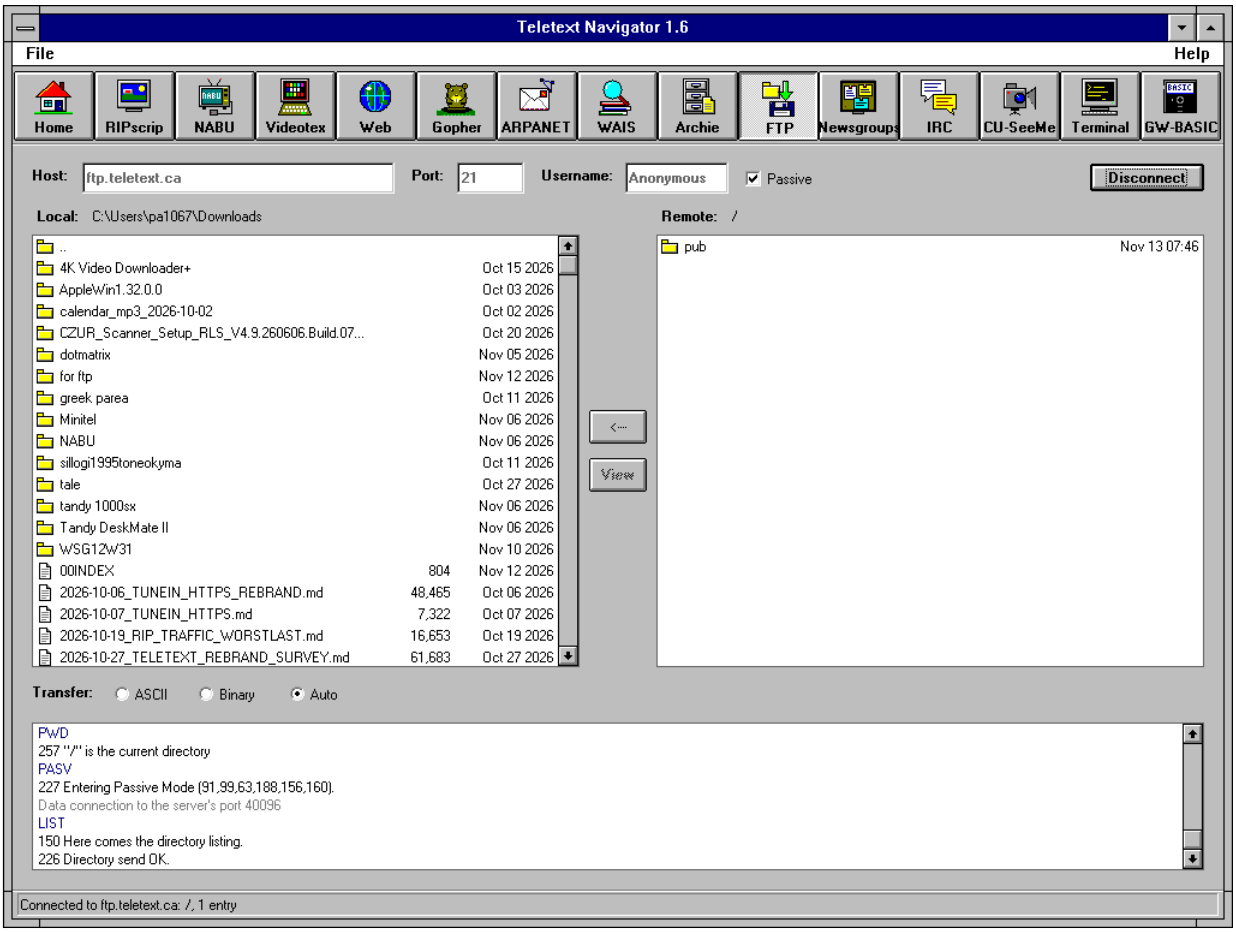



**Figure 18** The FTP service connected to ftp.teletext.ca.

## FTP Troubleshooting

### "Enter a host name." or "Port must be 1 to 65535."

Correct the Host or Port box.

### "Cannot connect: host" ... "not found."

The host name could not be found. Check its spelling.

### "Cannot connect to" the host ... "timed out" or "connection refused or unreachable"

The server did not accept a connection on that port. Check the port (21 for most servers).

**"Anonymous login refused"**
The server does not allow anonymous logins. The FTP service can log in only as anonymous.

**"Cannot open the data connection" or "Listing failed"**
Listings and files travel on a second connection. If Passive is cleared, check it and try again; if it is checked, try clearing it.

**"Transfer failed" ... "nothing kept."**
The download did not complete, and the partial file was deleted. Start it again.

**"The server ended the idle session" or "The server closed the idle connection. Disconnected."**
The server ends sessions that are idle. Connect again.

**"Not opened:" ... "is not a type View opens."**
View opens only text and PDF files. Download the file instead.

**"Could not open" ... "no viewer is set up"**
Windows has no program for that kind of file.

**"Refused to send" in the log**
The FTP service sends only the commands needed to list and download. Commands that would change a server are never sent.

CHAPTER 14



# Newsgroups

### About Netnews and NNTP

*NNTP*, the Network News Transfer Protocol, was one of the defining technologies of online news before the Web. For years, *Netnews* newsgroups and newsreader programs were a main way to read, follow and keep the information published on the network.

The Teletext runs its own news server, which carries its current news, weather bulletins and official alerts into this historic medium, in its own newsgroups:

- teletext.montreal.news: the Montreal and Canadian daily news.
- teletext.montreal.weather: Montreal conditions, the forecast and the weather briefing.
- teletext.montreal.alerts: official public alerts.
- teletext.montreal.traffic: Montreal road-work notices.
- teletext.montreal.numbers: markets and numbers (lottery results, gas prices, rates and currencies).

Anyone may read every group without logging in. Publicly reachable independent news servers of this kind appear to number only in the tens in the world.

### Newsgroups in Navigator

The Newsgroups service is a newsreader. It lists the groups of a news server, the articles of a group and the text of an article. It only reads: it cannot post an article, and it does not log in. It is set up for news.teletext.ca, port 119.

Connecting to a News Server . . . . . 46
Newsgroups Settings . . . . . 47
Reading Newsgroups . . . . . 47
Newsgroups Troubleshooting . . . . . 48

## Connecting to a News Server

1. Check the **Server** box (news.teletext.ca) and the **Port** box (119).
2. Choose **Connect**.

The status bar shows "Connecting to" the server, then "Reading group list...", then "Groups: Done" and the number of groups. The Newsgroups list fills, sorted by name, with each group's description under its name.

Choosing Connect clears all three panes. The server must answer within 20 seconds, and each reply within 30.

To disconnect, choose **Disconnect**. If nothing is being read, Navigator sends QUIT and shows the server's goodbye. If a list or an article is still arriving, it stops at once. What is on the screen stays there until you connect again.

## Newsgroups Settings

The only settings are the **Server** and **Port** boxes, which can be changed while you are disconnected. The usual NNTP port is 119.

The Newsgroups service connects without encryption and without a login, and remembers nothing after Navigator closes. It does not keep track of which articles you have read.

## Reading Newsgroups

The window has three panes:

- **Newsgroups**: the groups on the server. The number of articles appears once the group has been opened.
- **Articles**: the articles of the chosen group, newest first, with Subject, From, Date and Lines. Articles are listed one by one, not grouped into threads.
- **Article**: the text of the chosen article, with its From, Newsgroups, Subject and Date lines first.

### Choosing

- Click a group or an article to open it.
- Or move with the arrow keys, PAGE UP, PAGE DOWN, HOME and END; the row opens when you stop for a moment. ENTER opens it at once.
- TAB and SHIFT+TAB move between the three panes.

When a group opens, the status bar shows "Reading the overview of" the group, with the progress bar, then "Overview: Done" and the number of articles. Articles that are no longer on the server are left out.

### Reading

In the Article pane, use the arrow keys, PAGE UP, PAGE DOWN, HOME, END, the scroll bars or the mouse wheel. The text is shown in a fixed-pitch font, as the author wrote it.

Parts of an article that are not text, such as attachments, are named in brackets but not opened. An article longer than 8 megabytes is cut, and the status bar says so.

### Copying

Drag across the article's text to select it, then press CTRL+C, or click with the right mouse button and choose Copy. CTRL+A selects the whole article. The Newsgroups service cannot save an article.

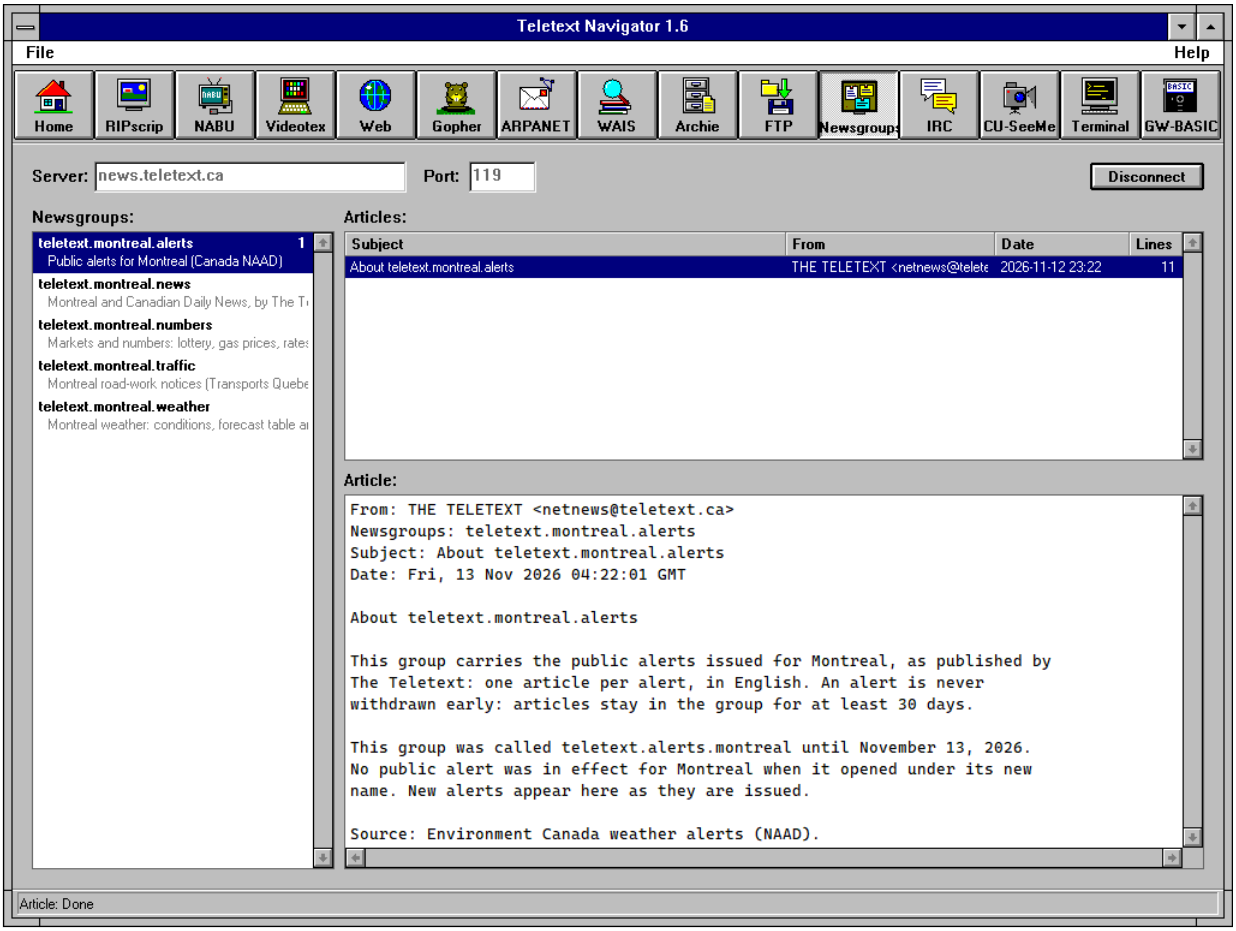



**Figure 19** The Newsgroups service reading an article of The Teletext's news server.

## Newsgroups Troubleshooting

**"Enter a server name." or "Port must be 1 to 65535."**
Correct the box, then choose Connect.

**"Cannot connect: host" ... "not found."**
The server's name could not be found. Check its spelling.

**"Cannot connect to" the server ... "timed out" or "connection refused or unreachable"**
The server did not answer on that port. The usual port is 119.

**"The server turned the connection away"**
The server refused this connection; its message says why.

**"Reading" a group "is not permitted"**
The server does not let anonymous readers open that group.

**"This article is no longer on the server"**
It has expired or been removed since the list was read. It is taken off the list.

**"The server ended the idle session" or "The server closed the idle connection. Disconnected."**
Connect again.

CHAPTER 15



# IRC

### About IRC

*IRC*, Internet Relay Chat, appeared in 1988 as a system for live text conversation over the network, and it soon carried live news during world events. IRC reporting was widely used during the Gulf War of 1991, the Soviet coup attempt of 1991, the Moscow crisis of 1993 and the Northridge earthquake of 1994.

The Teletext uses IRC in that tradition, as a live, read-only news channel: #retro carries news, alerts and weather updates in plain text, for old and new clients alike. The broadcast program is the only one allowed to speak in it.

### IRC in Navigator

The IRC service shows the messages of one channel as they arrive. It only receives: there is no box to type in, and it sends the server nothing but what IRC requires to stay connected. It is set up for irc.teletext.ca, port 6667, channel #retro.


Connecting to an IRC Channel . . . . . 49
IRC Settings . . . . . 49
Reading the Channel . . . . . 50
IRC Troubleshooting . . . . . 51


## Connecting to an IRC Channel

1. Check the **Server** box (irc.teletext.ca), the **Port** box (6667) and the **Channel** box (retro; the # is already there).
2. Choose **Connect**.

The status bar shows "Connecting to" the server and channel, then "Connected to" the server, "joining" the channel, then "Connected to" the server and channel. The window shows the server's welcome, and then the channel's messages.

Navigator joins under a name of its own, mgtview- followed by four digits, chosen anew at each connection.

If the connection drops after you have joined, Navigator connects again by itself, waiting a little longer each time, up to 30 seconds. The session stays open when you choose another service.

To disconnect, choose **Disconnect**. The text stays in the window.

## IRC Settings

- **Server** and **Port**: the IRC server. The Teletext's is irc.teletext.ca, port 6667.
- **Channel**: the channel to join, without its #. The Teletext's is retro.

These can be changed while you are disconnected. The IRC service connects without encryption and remembers nothing after Navigator closes.

## Reading the Channel

Each message is shown on a line of its own, with the time it was sent (in your time zone) in brackets. Every other message has a light green background, so long messages are easy to tell apart. The sender's name is not shown.

Only the messages and notices sent to the channel are shown. People joining or leaving, name changes and topic changes are not.

The last 2000 lines are kept. The window follows new messages while you are at the bottom; scroll up to read earlier ones, and new messages will not move the text away from you. Use the scroll bar, the mouse wheel, UP ARROW, DOWN ARROW, PAGE UP, PAGE DOWN, HOME (the oldest line) and END (the newest).

To copy, drag across the text and press CTRL+C, or click with the right mouse button and choose Copy. CTRL+A selects everything.

Letters outside the plain ASCII set, such as accented or Greek letters, are shown as blanks, and IRC colour codes are not shown in colour.

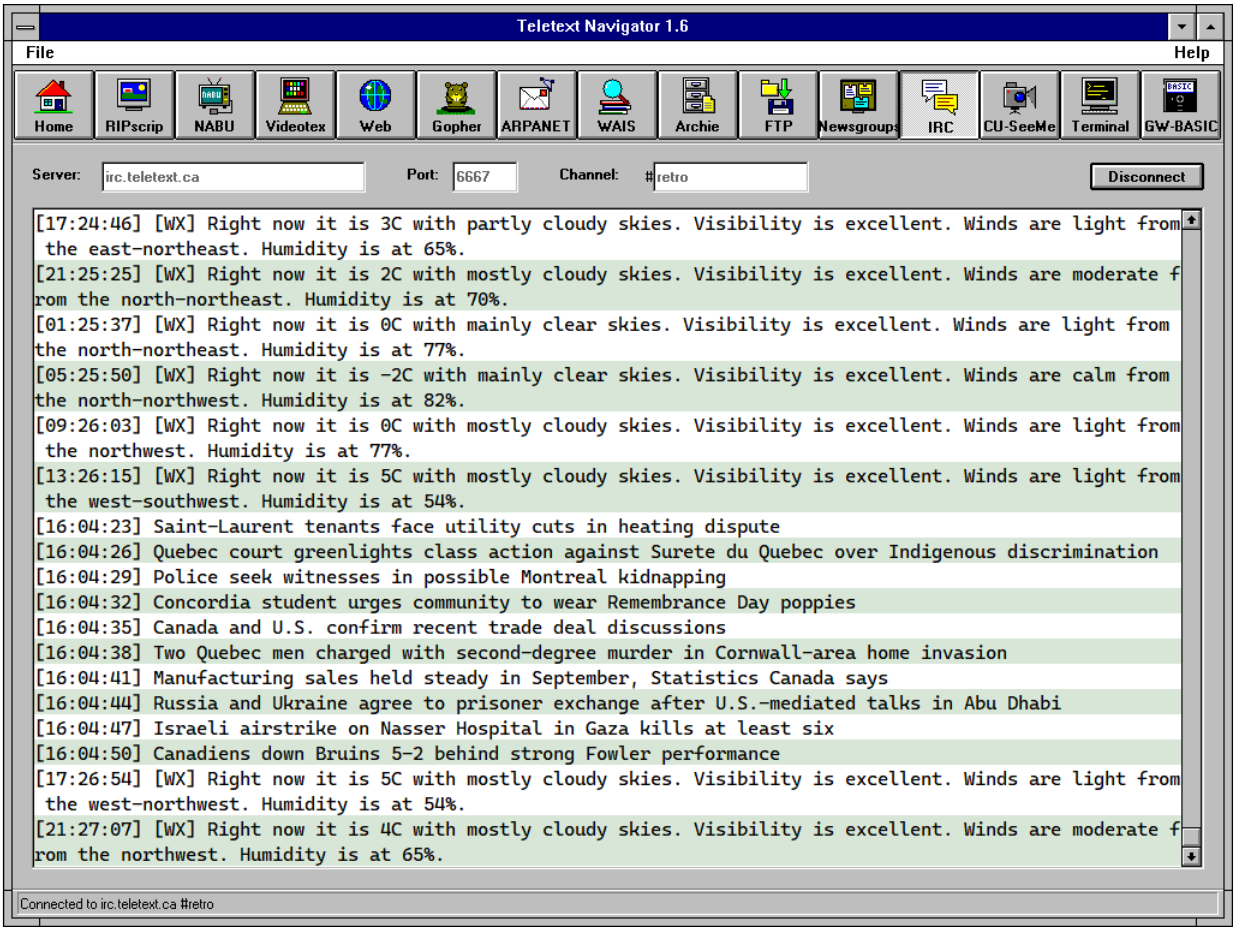



**Figure 20** The IRC service following #retro on irc.teletext.ca.

## IRC Troubleshooting

**"Server, port, and channel are required." or "Port must be 1-65535."**
Fill in the box, then choose Connect.

**"Connection failed: Could not resolve host"**
The server's name could not be found. Check its spelling.

**"Connection failed: Connection refused" or "Connection timed out"**
The server did not accept a connection on that port. The Teletext's server uses 6667.

**A message from the server in the window, beginning with ***
The server refused to let Navigator join the channel; its message says why. Check the channel's name.

**Nothing appears after joining**
The channel is quiet. On #retro, news and weather are sent at intervals.

CHAPTER 16



# CU-SeeMe

### About CU-SeeMe

In 1992 Cornell University built one of the Internet's first video systems: *CU-SeeMe*. When few people had ever seen live video travel over a network, it let computers around the world share grainy black-and-white pictures and sound in real time.

With a program called a *reflector*, one live feed could be sent to many viewers at once, and CU-SeeMe became something close to television. NASA used it to show space shuttle launches to the world, years before online video was ordinary. When the Oklahoma City bombing happened in 1995, the coverage of KOCO Channel 5 was carried by CU-SeeMe, one of the first times breaking news reached the world this way. Later that year ABC carried a live news program over the Internet with CU-SeeMe.

Teletext TV carries on that tradition: one live feed for anyone who wants to watch, with no chat room and nothing to send back. Its reflector runs the original Cornell software of 1995. To our knowledge, it is the only round-the-clock live television CU-SeeMe reflector running the original Cornell server still broadcasting anywhere in the world.

### CU-SeeMe in Navigator

The CU-SeeMe service shows the live picture and plays the sound, laid out like CU-SeeMe on a Macintosh of the time: a video window, a participant list, an audio window and two windows of statistics. It only watches and listens; it never sends. It is set up for cu-seeme.teletext.ca, which carries Teletext TV: 160 by 120 grey-scale video at about one frame a second, with sound in the Intel DVI format.


Connecting to a Reflector . . . . . 52
CU-SeeMe Settings . . . . . 53
Watching CU-SeeMe . . . . . 53
CU-SeeMe Troubleshooting . . . . . 54


## Connecting to a Reflector

1. Check the **Reflector** box (cu-seeme.teletext.ca) and the **Port** box (7648).
2. Check the **Audio Codec** list. Teletext TV uses Intel DVI.
3. Choose **Connect**.

The status line under the picture shows "Connecting...", then "Handshaking...", then "Receiving" with the frame rate, the sound rate and the codec. The picture replaces the test pattern, and Navigator appears among the Lurkers in the Participant List (a lurker watches without sending).

If nothing arrives for 8 seconds, the status line shows "Connection lost" and Navigator tries again by itself, waiting a little longer each time.

To disconnect, choose **Disconnect**; choosing it while connecting stops the attempt. Choosing another service also disconnects, so the sound does not play behind it. Choose Connect again when you return.

## CU-SeeMe Settings

- **Reflector** and **Port**: the reflector to watch. The Teletext's is cu-seeme.teletext.ca, port 7648.
- **Audio Codec**: how to decode the sound. **Intel DVI** (chosen at first) is what Teletext TV sends; **Delta-Mod** was used by other reflectors. Choose the one the reflector sends, or the sound will be noise.

These can be changed while you are disconnected. The volume and the mute button are in the Audio window. Nothing is remembered after Navigator closes.

## Watching CU-SeeMe

### The video window

Its title is the sender's name. The picture is shown four times its size when the window allows, or three or two times. The line under the picture shows the frame rate and the data rate, or DISCONNECTED, WAITING, TIMED OUT or PAUSED.

The buttons under the picture:

- The **eye** shows the test pattern instead of the picture (PAUSED). The sound keeps playing. Click it again to see the picture.
- The **speaker** mutes or restores the sound.
- The **two icons** at the right open and close the Transmission Statistics and Version and IP windows.

The microphone is dimmed: CU-SeeMe here never sends sound.

### The other windows

- **Participant List**: the people connected to the reflector, as Visible Users, Hidden Users and Lurkers.
- **Audio**: a speaker button (mute), the sound level and a volume slider.
- **Transmission Statistics**: how long you have been connected, frames and packets received, frame rate, data rate and the sound codec.
- **Version and IP**: the reflector's name and address, your address as the reflector sees it, and the sender's details.

The sound plays through your computer's speakers. The CU-SeeMe service is used with the mouse; no keys do anything in the picture area.

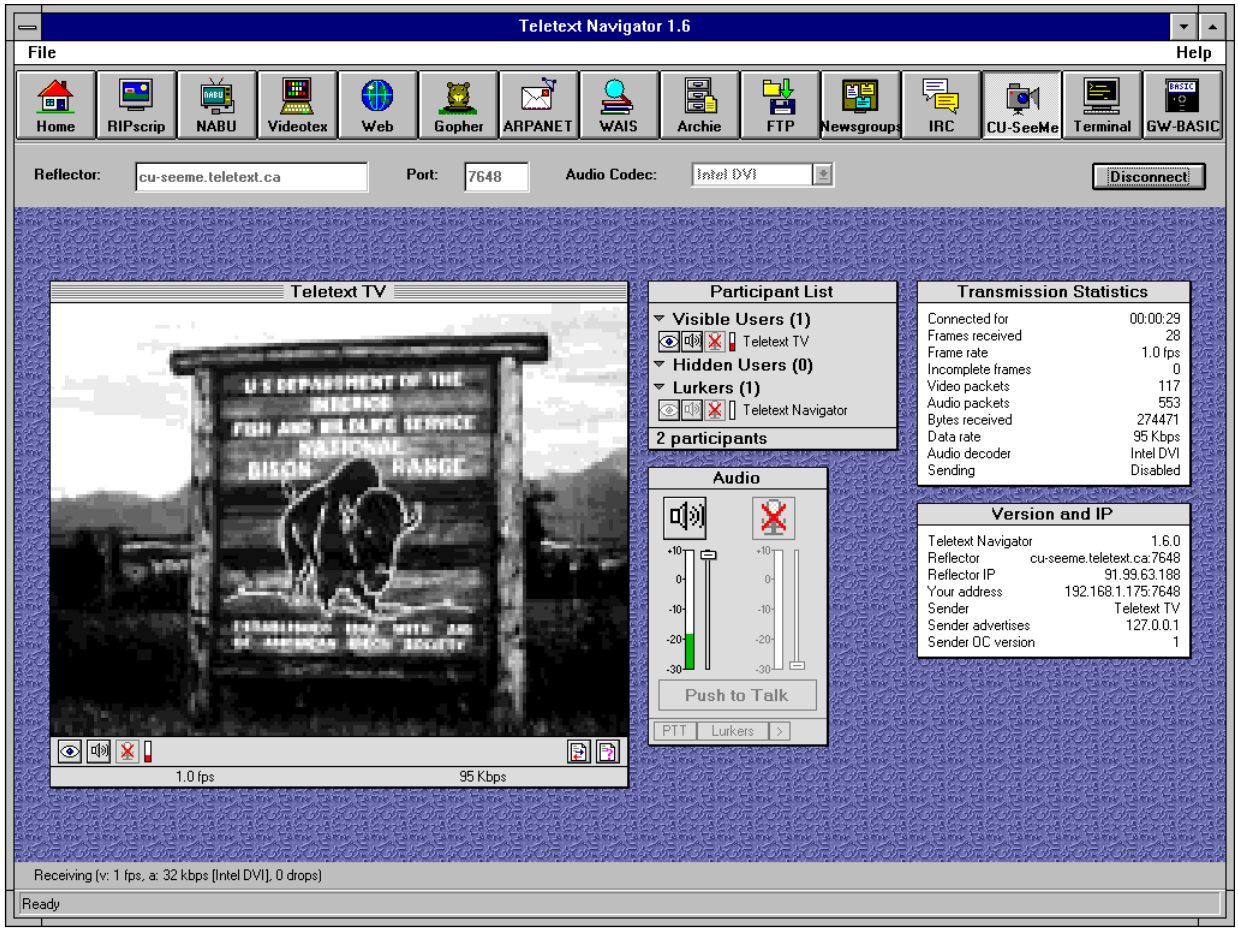



**Figure 21** CU-SeeMe receiving Teletext TV from cu-seeme.teletext.ca.

## CU-SeeMe Troubleshooting

### "Reflector and port are required." or "Port must be 1-65535."

Fill in the box, then choose Connect.

### "Connection lost - retrying"

Nothing has arrived for 8 seconds. Navigator keeps trying. If it never connects, check the reflector's name (a name that cannot be found gives the same message) and that your network allows UDP on port 7648.

### The sound is noise

Choose the other Audio Codec. Teletext TV sends Intel DVI.

### No sound at all

Check the speaker buttons and the volume slider, and your computer's sound. If Windows has no working sound device, CU-SeeMe plays no sound and gives no message.

### The picture stays on the test pattern

Check the line under the picture. PAUSED means the eye button is off; WAITING means no picture has arrived yet.

CHAPTER 17

# Terminal

### About Telnet, Finger and QOTD

*Telnet* is one of the original Internet protocols for using a remote, text-based service from a terminal. The Teletext runs a Telnet news service for quick, menu-driven reading: daily news, weather, traffic, markets and numbers, public alerts and service information. Public information services reached by Telnet appear to be extremely rare today, likely numbering only in the dozens in the world.

*Finger* is one of the earliest Internet services, from the late 1970s. It returns plain text, and early Internet users turned it into a bulletin: headlines, weather, storm updates, science and launch news. It declined because it could expose personal information; The Teletext's Finger service gives only public news, like a wire-service stub, with no personal lookup.

*QOTD*, the Quote of the Day (RFC 865, 1983), is the smallest service of all: opening a connection to port 17 is the request, and the reply is the whole message. The Teletext's sends a rotating "News of the Day": one headline summary and a Montreal weather line, with public alerts added when they are in force.

### The Terminal in Navigator

The Terminal is a text terminal with a small command prompt of its own. From it you can open a Telnet session, ask a Finger server for a bulletin, or read a QOTD message. Five buttons above it type the commands for The Teletext's services: **Telnet**, **Finger News**, **Finger Weather**, **QOTD** and **Archie** (the Archie server's own console). The terminal shows amber text on a black screen, and understands the *VT220* control sequences.


Connecting from the Terminal . . . . . 55
Terminal Commands . . . . . 56
Terminal Settings . . . . . 57
Using the Terminal . . . . . 57
Terminal Troubleshooting . . . . . 58


## Connecting from the Terminal

### With a shortcut button

Click a button at the top of the Terminal. Navigator types its command at the $ prompt and, one second later, presses ENTER for you:

- **Telnet**: telnet telnet.teletext.ca, The Teletext's menu-driven news service.
- **Finger News**: finger news@finger.teletext.ca, the headlines.
- **Finger Weather**: finger weather@finger.teletext.ca, the Montreal weather briefing.
- **QOTD**: nc qotd.teletext.ca 17, the news of the day.
- **Archie**: telnet archie.teletext.ca:2323, the Archie server's own console.

If you press a key during that second, the automatic ENTER is cancelled, so you can change the command first.

### By typing a command

At the $ prompt, type a command and press ENTER. See Terminal Commands (page 56).

### What you see

For Telnet, the Terminal prints "Connecting to" and the host and port; then the remote computer's screen appears. Finger and QOTD print their answer, and the server closes the connection.

When a session ends, the Terminal prints "Connection closed." and a new $ prompt, and the buttons can be used again. While a session is open the buttons are dimmed.

### Ending a Telnet session

The Terminal has no Disconnect button. End the session the way the remote service asks: choose its logout or quit command, or type quit at the Archie console. Everything you type during a session goes to the remote computer.

The session stays open if you choose another service.

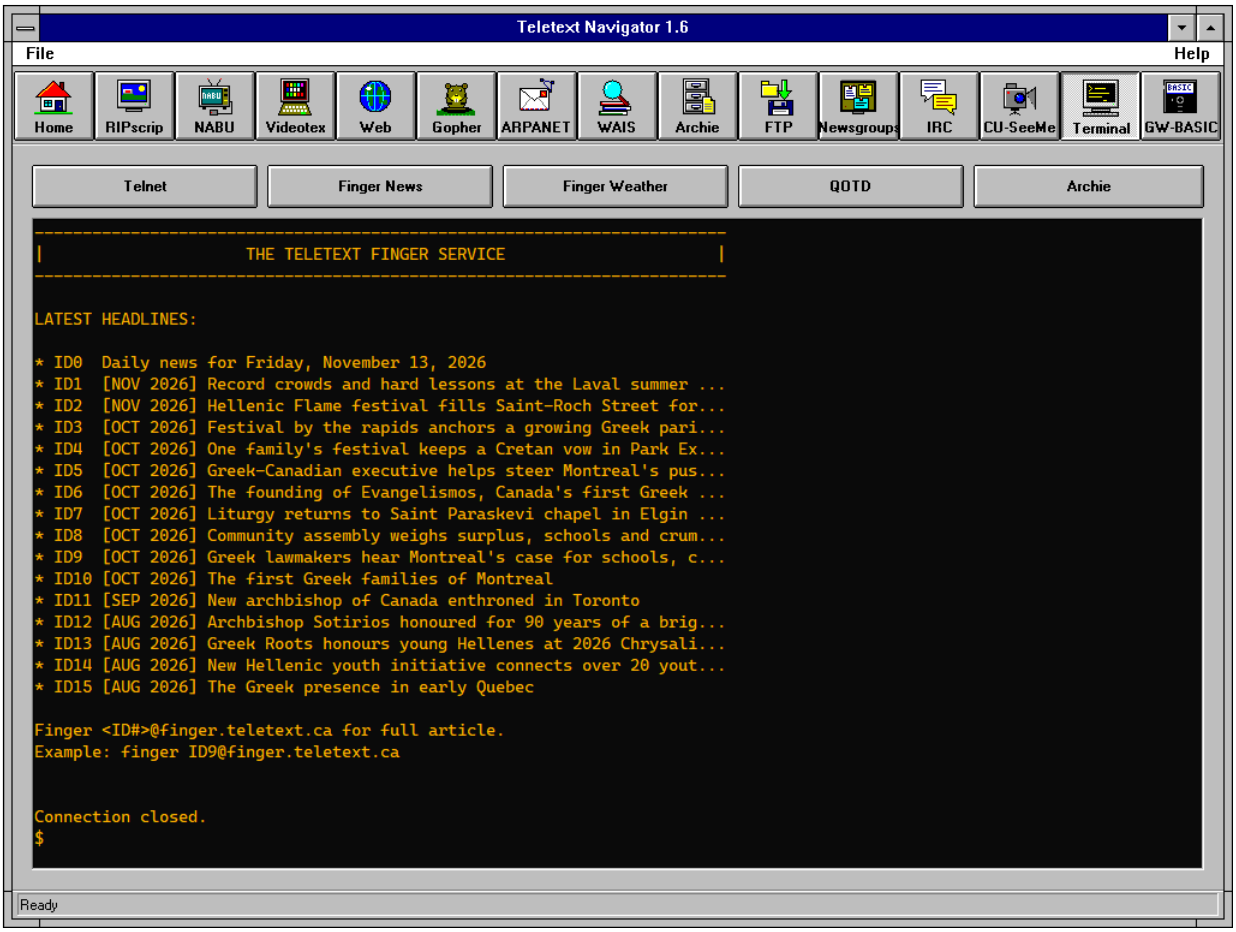



**Figure 22** The Terminal after the Finger News button.

## Terminal Commands

Type these at the $ prompt:

```
telnet host
telnet host:port
```

Opens a Telnet session. The port is 23 unless you give another.

```
finger name@host
```

Asks the Finger server on the host (port 79) about the name, and prints its answer.

```
nc host port
```

Opens a connection to the port and prints whatever the service sends, until it closes. Use it for QOTD (port 17).

```
clear
```
Clears the screen.
```
help
```
Lists the commands.
```
exit
quit
```
Closes the active connection.

### The Archie console

After telnet archie.teletext.ca:2323, the Archie server shows its own archie> prompt. Its commands include find (a substring search), exact (an exact file name), prog, help and quit. For a graphical search use the Archie (page 39) service.

## Terminal Settings

The Terminal has no settings, and remembers nothing after Navigator closes.

The screen is at least 80 columns by 24 rows and grows with the window. The text is drawn in Cascadia Mono, or Consolas if Cascadia Mono is not installed, at the largest size that fits. To the remote computer, the Terminal always reports a VT220 of 80 columns by 24 rows.

## Using the Terminal

### At the $ prompt

- ENTER runs the command. BACKSPACE deletes the character before the cursor.
- LEFT ARROW and RIGHT ARROW move in the line; HOME and END go to its ends.
- PAGE UP and PAGE DOWN scroll back through earlier lines.
- CTRL+L clears the screen and keeps the line you are typing.
- CTRL+C (with no text selected) abandons the line.

The prompt keeps no history of commands.

### In a Telnet session

- What you type is sent to the remote computer. ENTER sends a carriage return and line feed; BACKSPACE sends DEL.
- The arrow keys, HOME, END, PAGE UP and PAGE DOWN send the VT220 keys, for the remote program to use.
- CTRL+C (with no text selected) sends CTRL+C.
- The mouse wheel sends UP ARROW or DOWN ARROW three times a notch, which moves through most menus.
- Characters outside plain ASCII are not sent.

### Scrolling back

The Terminal keeps the last 1000 lines. When you are not in a Telnet session, PAGE UP, PAGE DOWN and the mouse wheel scroll back through them. New text returns you to the bottom.

### Copying and pasting

- Drag across text to select it. CTRL+A selects everything.
- CTRL+C copies the selection.
- CTRL+V pastes text at the prompt or into a Telnet session; each line break is sent as ENTER.
- Or click with the right mouse button and choose Copy, Paste or Select All.

### What the screen shows

The Terminal understands the VT220 sequences that clear the screen, move the cursor and erase lines. Colours and bold text are shown in plain amber, and the bell is silent. Characters outside plain ASCII appear as spaces.

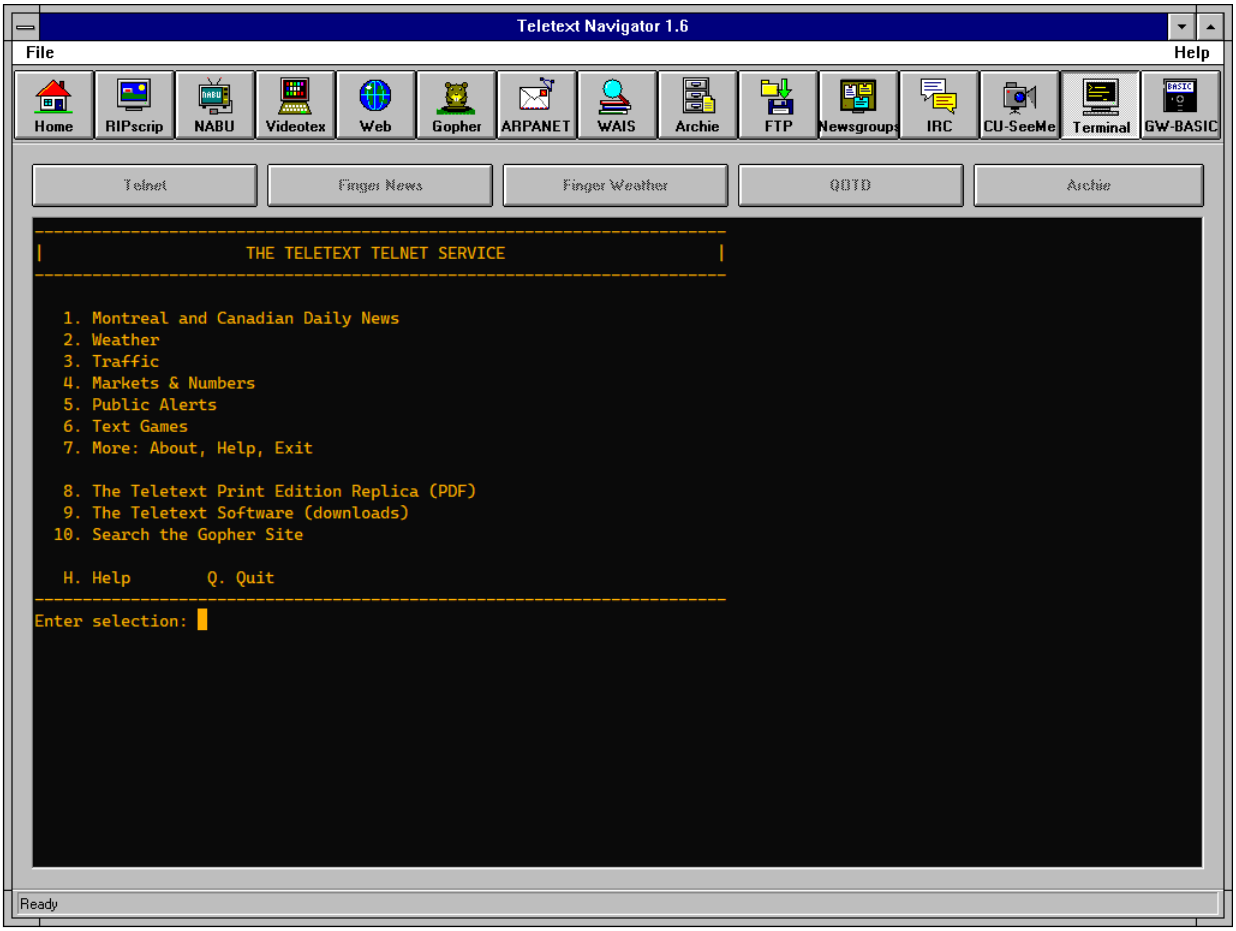



**Figure 23** A Telnet session with telnet.teletext.ca.

## Terminal Troubleshooting

### "command not found"

The prompt knows only the commands in Terminal Commands (page 56). Type help to list them.

### "usage:" followed by a command

The command needs more: for example telnet needs a host, finger needs name@host, and nc needs a host and a port.

### "nc: invalid port"

Ports run from 1 to 65535.

### "Error: Host not found"

The host name could not be found. Check its spelling.

### "Error: Connection refused" or "Error: Connection timed out"

The service did not accept a connection on that port within 20 seconds.

### "No active connection."

exit and quit close a connection; there is none open.

### The shortcut buttons stay dimmed

They are dimmed during a session. If you pressed a key during the second after clicking a button, the command was not run: press ENTER to run it, or BACKSPACE it away and run another command; the buttons return when that session ends.

### I cannot leave a Telnet session

Use the remote service's own logout or quit command. The Terminal has no Disconnect button.

CHAPTER 18




# GW-BASIC

### About GW-BASIC

GW-BASIC is the BASIC that Microsoft supplied with the IBM PC compatibles of the 1980s, from 1983. For a generation of home and office computer users it was the first programming language, and magazines printed program listings in it for readers to type in. Microsoft published its source code in 2020, under the MIT licence.

### GW-BASIC in Navigator

The GW-BASIC service runs Microsoft GW-BASIC on a PC built into Navigator: an 8088 processor at 4.77 MHz with a *CGA* display and a PC speaker. Write and run programs as on a real PC. Programs are saved in a folder of your Documents, where two programs are waiting to start with: RETRO649.BAS and DIALUP.BAS.

Starting GW-BASIC . . . . . 60
GW-BASIC Files and Settings . . . . . 60
Using GW-BASIC . . . . . 61
GW-BASIC Troubleshooting . . . . . 62

## Starting GW-BASIC

GW-BASIC has nothing to connect to. It starts when you choose its button, and shows its Ok prompt.

- **Restart** starts GW-BASIC again. The program in memory is lost; saved files are not touched.
- The SYSTEM command ends GW-BASIC. Press any key in the screen to start it again.

When you choose another service, GW-BASIC is paused, with its sound; when you return it carries on, with your program still in memory.

GW-BASIC does not warn you about a program you have not saved. SAVE it before you close Navigator.

## GW-BASIC Files and Settings

GW-BASIC has no settings.

### Where files are kept

To GW-BASIC, drive C: is the folder **Teletext BASIC** in your Documents. The line under the buttons shows where it is. LOAD, SAVE, FILES, KILL, NAME, OPEN, MKDIR, CHDIR and RMDIR work in that folder, and nothing outside it can be reached; drive letters are ignored.

Names follow the DOS rules: up to eight characters, a period and three more. They are stored in capitals, and longer names are cut. Files with long Windows names are not seen.

Choose **Open BASIC folder** to see the folder in Windows.

# Using GW-BASIC

## Keyboard

Click the screen to give it the keyboard. Keys are typed as on a US keyboard, whatever layout Windows uses.

- F1 to F10 are GW-BASIC's function keys, with SHIFT, CTRL and ALT. (In GW-BASIC, F1 is LIST.)
- ALT with a letter types a BASIC keyword.
- The arrow keys, HOME, END, PAGE UP, PAGE DOWN, INS and DEL edit as on the PC.
- CTRL+BREAK stops a running program. CTRL+SCROLL LOCK and CTRL+PAUSE do the same.

ALT+F4 still closes Navigator.

## Pasting a program

Copy a listing from another program, then choose **Paste** or press CTRL+V or SHIFT+INS. Navigator types it into GW-BASIC, line by line. Curly quotation marks and dashes become their plain ASCII forms; other characters outside ASCII are left out.

## Screen and sound

GW-BASIC's text screens and its CGA graphics screens are shown at 4:3. BEEP, SOUND and PLAY play as on a PC speaker.

Clicking the screen only gives it the keyboard; programs cannot read the mouse. Text cannot be copied out of the screen.

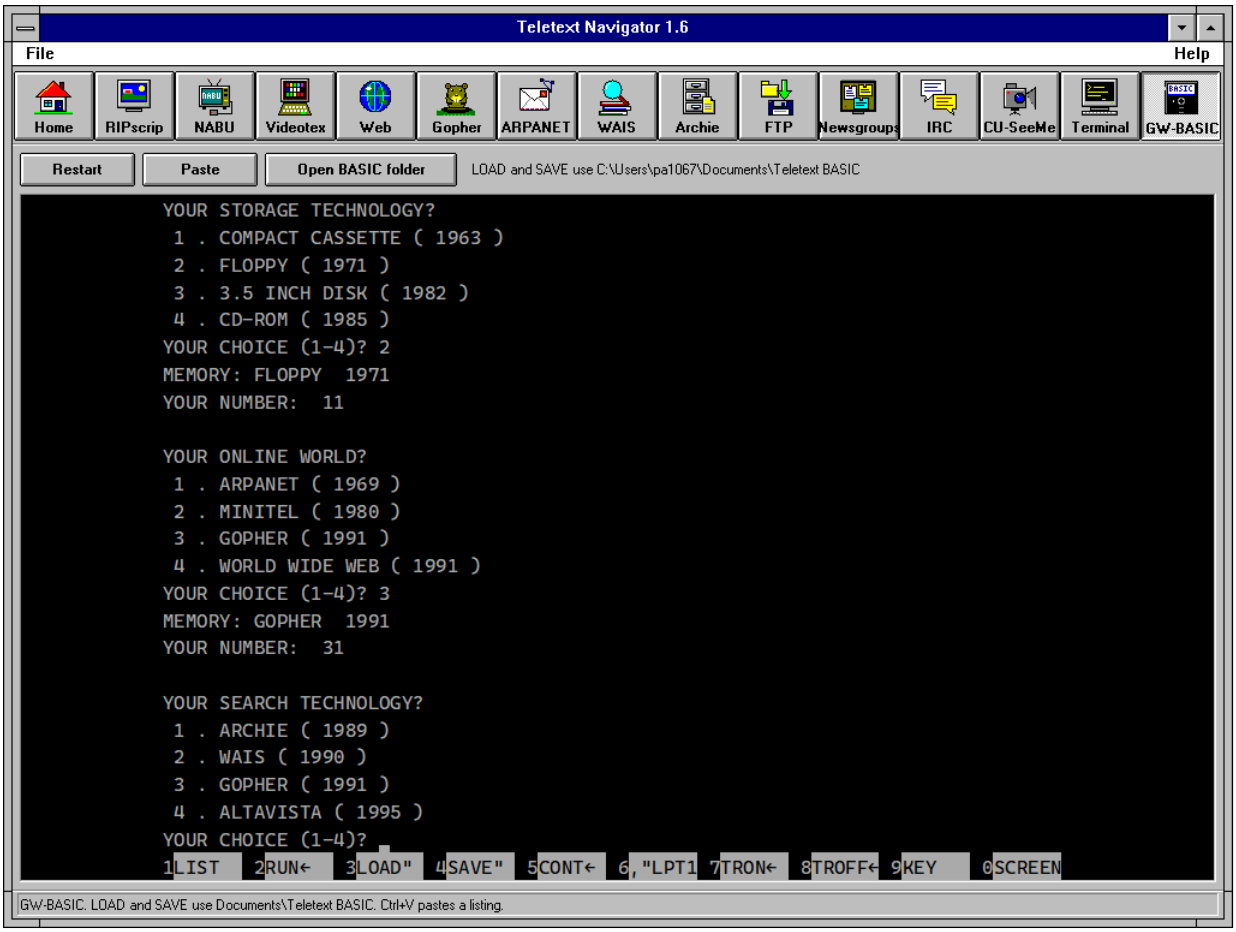



**Figure 24** GW-BASIC running RETRO649.BAS.

## GW-BASIC Troubleshooting

### "GW-BASIC could not start"

The reason follows. "the BASIC folder does not exist" or "could not create Documents\Teletext BASIC" means Navigator could not make the folder in your Documents; check that Documents can be written to.

### A file is not found

Only the Teletext BASIC folder can be reached, and names must follow the DOS 8.3 rules.

### GW-BASIC's own error messages

Messages such as Syntax error are GW-BASIC's. See a GW-BASIC manual.

### SHELL, the printer and the serial ports

This PC cannot run other programs with SHELL, and it has no printer and no serial ports. Programs that need them will not work as written.

### The screen shows only text when a program draws

Only the CGA screen modes are shown. Programs written for EGA or VGA screens fall back to text.

### F1 does not open Help

In GW-BASIC, F1 belongs to the BASIC screen. Choose Contents from the Help menu instead.

CHAPTER 19

# Using Help

Teletext Navigator's Help is a Windows Help file, TELETEXT.HLP, shown by **The Teletext WinHelp** (TTWINHLP.EXE). It holds the same information as this manual, ready on the screen while you work.

To open Help:

- Choose **Contents** from the Help menu to start at the Contents.
- Choose **Search for Help on** to find a topic by keyword.
- Press F1 for the topic of the service you are using. In GW-BASIC and Videotex, F1 belongs to the screen of the service; see Getting Help (page 10).
- Choose **How to Use Help** to learn about the Help window itself.

One Help window serves every request: when you press F1 in another service, the same window moves to that service's topic. When you close Navigator, Help closes with it.

The Teletext WinHelp is the WinHelp viewer of the Wine project, made into a native Windows program by The Teletext, with the look of WinHelp 3.1. It is free software under the GNU Lesser General Public License, version 2.1 or later; its source code is at https://github.com/TheTeletext/WinHelp. It also opens the Help files of Windows 3.0 and Windows 3.1 programs.

The rest of this chapter is the viewer's own How to Use Help.

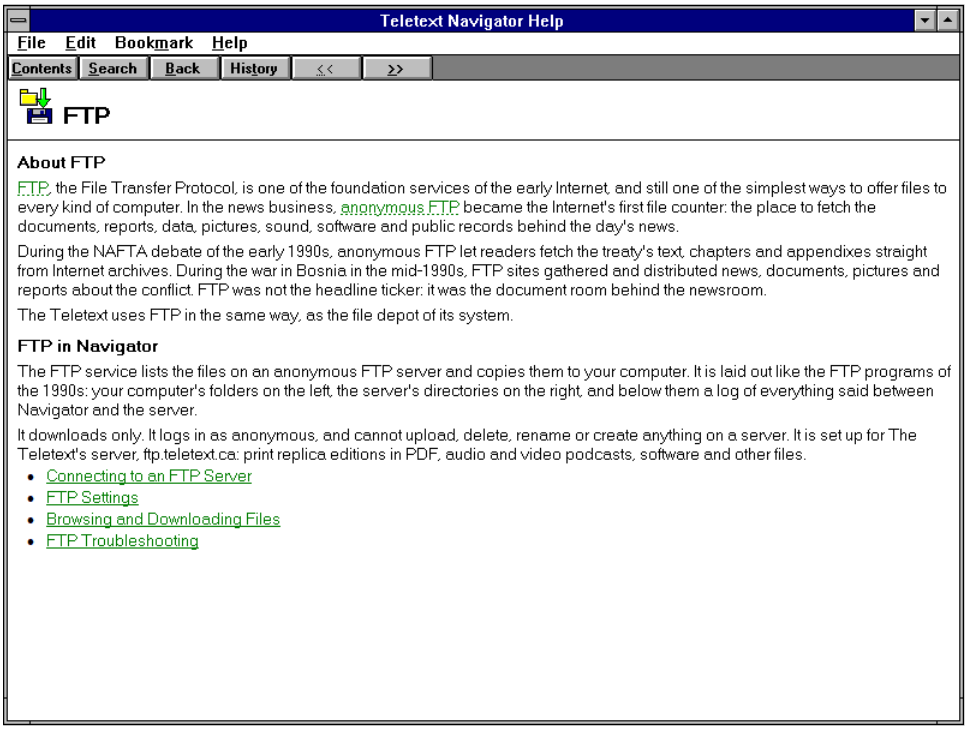



**Figure 25** The Help topic opened by F1 in the FTP service.

## How to Use Help

Help shows information about a program on your screen, while you work. Choose a topic to learn more.

### Moving around in Help

Jumps and Definitions . . . . . 64
The Help Buttons . . . . . 65
Searching for a Topic . . . . . 65
Going Back to Earlier Topics . . . . . 66

### Working with Help

The Help Window . . . . . 66
Copying Help Text . . . . . 66
Keeping Help on Top . . . . . 66
Opening Another Help File . . . . . 66
Help Menu Commands . . . . . 66

### About this program

About The Teletext WinHelp . . . . . 67

## Jumps and Definitions

Words in a Help topic that are shown in color and underlined lead you further. In most Help files they are green.

### Jumps

A word or phrase with a solid underline is a **jump**. Click it to go to the topic it names. The pointer becomes a hand when it is over a jump.

### Definitions

A word with a dotted underline has a **definition**. Click it and the definition appears in a small window over the topic. Click anywhere, or press any key, to close it.

Some Help files also use pictures as jumps or definitions. The pointer becomes a hand over them too.

To return to the topic you came from, choose the Back button.

## The Help Buttons

The buttons under the menu bar:

- **Contents**: shows the Help file's contents, the topic it starts with.
- **Search**: finds topics by keyword. See Searching for a Topic (page 65).
- **Back**: returns to the topic you saw before this one. It is dimmed at the first topic.
- **History**: lists the topics you have seen. See Going Back to Earlier Topics (page 66).
- << and >>: move to the previous or next topic in a series of related topics, called a browse sequence. A button is dimmed at the end of the series.

Each button has an underlined letter: hold down ALT and press the letter to choose the button. Some Help files add buttons of their own.

## Searching for a Topic

To find a topic by keyword:

1. Choose the **Search** button. The Search dialog box opens.
2. Type a word in the box, or choose one from the list. As you type, the list moves to the first keyword that begins with what you have typed.
3. Choose **Show Topics**. The topics that have this keyword appear in the lower list.
4. Choose a topic, then choose **Go To**.

You can also double-click a keyword instead of choosing Show Topics, and double-click a topic instead of choosing Go To. After Show Topics, the Cancel button reads Close.

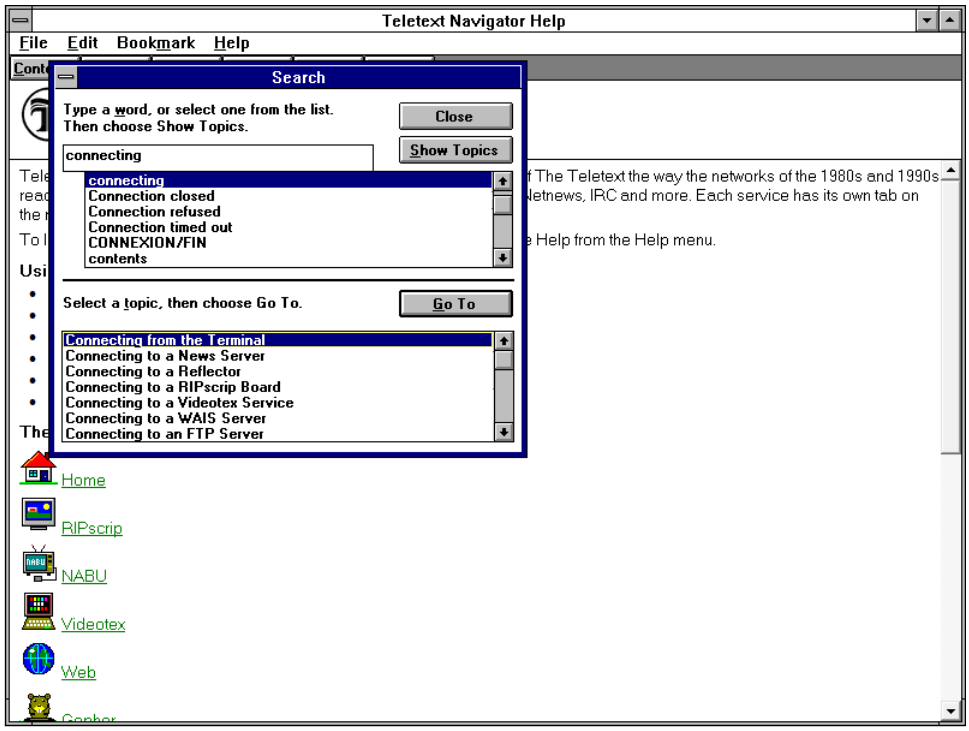



**Figure 26** The Search dialog box: the topics with the keyword "connecting".

## Going Back to Earlier Topics

### Back

Choose **Back** to return to the topic you saw before this one. Choose it again to go further back.

### History

Choose **History** to open the Windows Help History window, which lists the last topics you have seen, the most recent first. Topics from another Help file are shown with that file's name.

To go to a topic, double-click it, or select it with the arrow keys and press ENTER. The History window stays open until you close it.

## The Help Window

### Scrolling

Use the scroll bar, the mouse wheel, or the PAGE UP, PAGE DOWN, UP ARROW and DOWN ARROW keys.

In many Help files, the title of the topic stays at the top of the window, in a band of its own, while the rest of the topic scrolls under it.

### Size and position

Help remembers the size and position of its window for the next time.

### Other windows

Some Help files show certain topics in a window of their own, such as a glossary. These windows have no menu bar and no buttons; close them with their Control-menu box.

### Closing Help

Choose Exit from the File menu, or close the window with its Control-menu box.

## Copying Help Text

To copy text from a Help topic:

1. Drag across the text with the mouse to select it.
2. Choose **Copy** from the Edit menu, or press CTRL+C.

The text is placed on the Clipboard, and you can paste it into another program.

## Keeping Help on Top

To keep the Help window in front of the program you are using, choose **Always on Top** from the Help menu. A check mark appears beside the command. Choose it again to turn it off.

## Opening Another Help File

1. Choose **Open** from the File menu.
2. Choose a Help file (a file whose name ends in .HLP), then choose Open.

Help reads the Help files of Windows 3.0 and Windows 3.1.

## Help Menu Commands

### File menu

- **Open**: opens another Help file. See Opening Another Help File (page 66).
- **Exit**: closes Help.

### Edit menu

- **Copy**: copies the selected text. See Copying Help Text (page 66).

### Help menu

- **How to Use Help**: shows this Help. You can also press F1 in any Help window.
- **Always on Top**: keeps Help in front of other windows. See Keeping Help on Top (page 66).
- **About Help**: shows the version of this program, its licence and the address of its source code.

### Not yet available

Print Topic, Print Setup, Annotate and Define a bookmark are on the menus, as in Windows 3.1, but they do not work yet in this version.

## About The Teletext WinHelp

The Teletext WinHelp (TTWINHLP.EXE) shows the Help files of Windows 3.0 and Windows 3.1 on today's Windows, where Microsoft no longer provides a Help program for them.

It is based on the WinHelp viewer of the Wine project, and is free software under the GNU Lesser General Public License, version 2.1 or later. Its source code is at https://github.com/TheTeletext/WinHelp.

Choose About Help from the Help menu for its version and credits.

# Glossary

**anonymous FTP**

A way of offering files by FTP to anyone. You log in with the user name "anonymous" and need no account of your own. Navigator's FTP and Archie services always log in this way.

**Archie**

The first Internet search engine, created at McGill University in Montreal in 1989. An Archie server keeps an index of the files on anonymous FTP sites and tells you which host and directory hold a file.

**ARDP**

The Asynchronous Reliable Delivery Protocol of the Prospero system. Archie clients use it to send a request to the server over UDP, and to receive the answer in numbered packets.

**ASCII mode**

An FTP transfer mode for text files. The server sends lines of text, and line endings are adjusted for the receiving computer. Use binary mode for everything else.

**binary mode**

An FTP transfer mode in which every byte of the file is sent unchanged. Programs, archives, pictures, sound and PDF files must be sent in binary mode.

**CGA**

The Color Graphics Adapter, IBM's first colour display for the PC (1981): 80 by 25 or 40 by 25 text, and graphics up to 640 by 200 pixels.

**codec**

Short for coder-decoder: the method used to turn sound into data and back. CU-SeeMe used several; the Teletext TV reflector sends Intel DVI.

**CU-SeeMe**

A program written at Cornell University in 1992 to send live video and sound over the Internet. A reflector passes one sender's picture and sound on to many viewers.

**DNS name**

The name of a computer on the Internet, such as news.teletext.ca. The Domain Name System turns the name into the computer's numeric address.

**EGA**

The Enhanced Graphics Adapter (IBM, 1984): 640 by 350 pixels in 16 colours. RIPscrip screens are drawn at this size.

**Finger**

One of the earliest Internet services. You ask a Finger server about a name, as in news@finger.teletext.ca, and it answers with text, then closes the connection. Finger uses port 79.

**FTP**

The File Transfer Protocol, one of the oldest Internet services, used to list the files on a server and copy them to your computer. FTP servers usually listen on port 21.

**FTP-Mail**

Electronic mail sent with the MAIL command of the File Transfer Protocol, as on the ARPANET before mail had a protocol of its own (RFC 765).

**Gopher**

A menu-based way of publishing documents on the Internet, created at the University of Minnesota in 1991 (RFC 1436). Each menu item leads to another menu, a document, a search or a file.

**host**

A computer on the network that offers a service. In Navigator you type the host's DNS name, such as wais.teletext.ca.

**IRC**

Internet Relay Chat, created in 1988: text conversations in channels whose names begin with #. Servers usually listen on port 6667.

**Minitel**

The French videotex terminal of the Teletel service, introduced in 1982: 40 columns by 25 rows, with its own keyboard of function keys such as ENVOI and SOMMAIRE.

**NABU**

The NABU Network, a Canadian home computer service delivered by cable television from 1983. Programs were sent down the cable to the NABU Personal Computer, a Z80 machine, which ran them.

**NAPLPS**

The North American Presentation Level Protocol Syntax, the graphics standard that grew out of Canada's Telidon videotex. Pictures are sent as drawing commands, not as pixels.

**Netnews**

The worldwide system of newsgroups, in which articles are copied from server to server. Newsreaders fetch articles from a news server with NNTP.

**NNTP**

The Network News Transfer Protocol (RFC 977, 1986; now RFC 3977), used by newsreaders to list newsgroups and read their articles. News servers usually listen on port 119.

**passive mode**

An FTP setting in which your computer opens the data connection to the server, instead of the server opening it to your computer. Passive mode works through most firewalls and home routers.

**port**

A number from 1 to 65535 that, with the host name, says which service on the host to connect to. Each service has a usual port: 21 for FTP, 23 for Telnet, 70 for Gopher, 119 for NNTP, 210 for WAIS.

**QOTD**

The Quote of the Day service (RFC 865, 1983). A connection to port 17 returns one short message and closes. The Teletext sends a news bulletin of the day.

**reflector**

A CU-SeeMe server that receives a sender's video and sound and passes them on to every connected viewer.

**RIPscrip**

The Remote Imaging Protocol, developed by TeleGrafix Communications in the early 1990s. A RIPscrip bulletin board sends short text commands that the terminal draws as lines, shapes, text and buttons, so graphical screens could travel over a modem.

**selector**

In Gopher, the string that names one item on a server. Navigator shows the full address of an item, as gopher://host/selector, in its Selector box.

**Telnet**

The original Internet protocol for using a remote computer from a terminal (RFC 854). Telnet servers usually listen on port 23.

**VT220**

A text terminal made by Digital Equipment Corporation in 1983. Its control sequences, which move the cursor and erase the screen, became a standard that most terminal programs understand.

**WAIS**

Wide Area Information Servers, developed from 1989 by Thinking Machines, Apple Computer and Dow Jones. You send words to a WAIS server and it returns the documents of a database ranked by how well they match.

**Z39.50-1988**

The American national standard for information retrieval, which WAIS used to carry its searches. Navigator speaks Z39.50-1988 (Version 1), the version of the original WAIS servers.

**Zmodem**

A file transfer protocol written by Chuck Forsberg in 1986 and widely used by bulletin boards. The board sends the file in checked blocks, and the terminal saves it.

# Index


**#**

#retro, 49

**A**

About, 7, 67
Active Channel, 23
Always on Top, 66
anonymous, 42
anonymous FTP, 42
anonymous login refused, 44
Archie, 39, 40, 41
Archie console, 55
Archie server, 39
ARPANET, 31, 32, 33
articles list, 47
ASCII, 43
attachments, 47
audio codec, 53
Auto, 43

**B**

Back, 66
Back button, 65
BASIC, 60
BBS, 12
Binary, 43
Bookmark menu, 66
browse buttons, 65
browser, 22
bulletin board, 12
buttons, 13, 65

**C**

cancelling a transfer, 43
capital letters, 20
CAPITALS only, 20
channel, 16, 17, 49
Check for Updates, 9
choosing a service, 7, 11
Clear, 32
clear, 56
clipboard, 66
colours, 57
command not found, 58
command prompt, 55
commands, 56
Connect button, 12, 16, 19, 31, 34, 42, 46, 49, 52
connecting, 12, 16, 19, 22, 28, 31, 34, 39, 42, 46, 49, 52, 55
Connection closed, 58
Connection refused, 14
Connection timed out, 14
CONNEXION/FIN, 19
contents, 64
Contents button, 65
context help, 10
conventions, 3
Copy, 66
copying, 66
copying text, 23, 29, 47, 50, 57
CU-SeeMe, 52, 53, 54

**D**

database, 34, 35
database not found, 37
definitions, 64
Delta-Mod, 53
Directories, 40
directories, 43
Disconnect, 12
Doc #, 35
downloading files, 13, 40, 42, 43
Downloads folder, 13, 43

**E**

earlier topics, 66
Edit menu, 66
emulator, 16
ending a session, 55
ENVOI, 20
errors, 14, 18, 21, 26, 30, 33, 37, 41, 44, 48, 51, 54, 58, 62
Exact first, 40
Exit, 7
exit, 56

**F**

F1, 10
File menu, 7, 66
file transfer, 42
Files, 40
files, 60
finding a topic, 65
Finger, 55
finger command, 56
firmware, 18
Flash, 22, 23, 26
font, 57
forms, 26
FTP, 42, 43, 44
FTP archives, 39
FTP-Mail, 31, 33
ftp:// addresses, 42
full-text search, 34
function keys, 61

**G**

Go, 22, 28
Go To, 65
Gopher, 28, 29, 30
Gopherspace, 28

graphical terminal, 12
GW-BASIC, 60, 61, 62

**H**
help, 10, 56
Help, 63, 64
Help Compiler, 2
Help key, 10
Help menu, 7, 66
Help viewer, 63
Help window, 66
history, 23, 29
History, 66
History button, 65
Home, 11, 22, 28
host, 12, 13, 42
Host not found, 14
Hosts, 40
hotkeys, 13
hotspots, 64
How to Use Help, 64
HTML, 22
HTTPS, 26

**I**
idle session, 44
installed files, 4
installer, 4
installing, 4
installing an update, 9
Intel DVI, 53
Internet Relay Chat, 49
introduction, 1
IRC, 49, 50, 51

**J**
jumps, 64

**K**
keeping Help visible, 66
keyboard, 17, 20, 57, 61
keys, 3
keywords, 65

**L**
licence, 67
links, 23
Live TV, 52
LOAD, 60
local folder, 43
login, 42
look and feel, 2

**M**
mail, 31
mailbox, 31
manual, 4
menu items, 29
menus, 7, 28, 66
message size, 32
messages, 9
Microsoft BASIC, 60
Minitel, 19, 21
Minitel keys, 20
Mode, 17
Monochrome, 20
mouse, 13
mouse wheel, 57
MSI, 4
mute, 53

**N**
NABU, 16, 17, 18
NABU keys, 17
NABU Network, 16
NAPLPS, 19, 21
NAPLPS keys, 20
native program, 2
Navigator window, 6
nc command, 56
Netnews, 46
new version, 4, 9
Newsgroups, 46, 47, 48
newsreader, 46
nickname, 49
NNTP, 46, 48
No channel connection, 18
no picture, 54
no response, 41
no sound, 54

**O**
on top, 66
Open, 66
Open BASIC folder, 60
opening a Help file, 66
opening a menu, 28
opening a page, 22
overview, 1, 12, 16, 19, 22, 28, 31, 34, 39, 42, 46, 49, 52, 55, 60

**P**
participant list, 53
Passive, 43
passive mode, 44
pasting, 57
pasting a program, 61
pictures, 29
popups, 64
port, 12, 13, 31, 34, 35, 42, 46, 47, 49, 52
printer, 62
programming, 60
progress, 9

**Q**
QOTD, 55

**R**
reading a document, 35
reading articles, 46, 47
reading messages, 50
reflector, 52
regular expression, 40
removing, 4
requirements, 3
resizing the window, 6
Restart, 60
returning to a topic, 66
ribbon, 7
RIPscrip, 12, 13, 14

**S**
SAVE, 60
Save As, 40
saving files, 29
Scanlines, 20
score, 35
screen modes, 61
screen size, 57
scrolling, 23, 50, 57, 66
Search, 65
Search button, 39, 65
search type, 40
search words, 35

searching, 29, 34, 35, 39
searching for files, 39
secondary windows, 66
selecting text, 66
Selector, 28
Send, 32
sending a message, 31
serial port, 17, 18
server, 31, 34, 35, 46, 47, 49
settings, 13, 17, 20, 23, 29, 31, 35, 40, 43, 47, 49, 53, 57
setup, 4
SHELL, 62
shortcut buttons, 55
shortcuts, 4
Show Topics, 65
skin, 2
SOMMAIRE, 20
sound, 17, 23, 29, 53, 61
source code, 67
Start Menu, 4
starting, 60
starting Navigator, 4, 11
starting the NABU, 16
statistics, 53
status bar, 9
status line, 9
Stop, 30
substring, 40
switching services, 7
SYSTEM, 60
system requirements, 3

**T**

tabs, 7
Teletext BASIC folder, 60
Teletext Navigator, 1
Teletext TV, 52
Telidon, 19
Telnet, 12, 55
telnet command, 56
Terminal, 55, 56, 57, 58
The Teletext WinHelp, 2, 63, 67
title band, 66
title bar, 6
To, 31
transfer mode, 43
troubleshooting, 14, 18, 21, 26, 30, 33, 37, 41, 44, 48, 51, 54, 58, 62
TTWINHLP, 63
typographic conventions, 3

**U**

underlined words, 64
uninstalling, 4
updates, 9
updating, 4
URL, 22
using Help, 64

**V**

video, 52
video window, 53
Videotex, 19, 20, 21
View, 35, 43
volume, 53

**W**

WAIS, 34, 35, 37
Web, 22, 23, 26
window, 6
window size, 66
Windows 10, 3
Windows 11, 2, 3
Windows 3.1 look, 2
Wine, 67
WinHelp, 2
World Wide Web, 22
writing a message, 32

**X**

x64, 2

**Z**

Z39.50, 34
Zmodem, 13